

FUNCTIONS

LEVEL-V

1. $f: R \rightarrow R, f(x) = x^3 - 3x^2 + 6x - 5$ is
A) one-one and onto
B) one-one and into
C) onto but not one-one
D) neither one-one nor onto
2. Let $f: R \rightarrow R$ be a function defined by
$$f(x) = \frac{x^2 + 2x + 5}{x^2 + x + 1}$$
 is
A) one-one and into B) one-one and onto
C) many-one and onto D) many-one and into
3. $f: R \rightarrow R, f(x) = \frac{x^2 + 4x + 30}{x^2 - 8x + 18}$ is
A) one-one & onto B) many-one & onto
C) one-one & into D) many-one & into
4. $f: R \rightarrow R$ is a function defined by
$$f(x) = \frac{e^{|x|} - e^{-x}}{e^x + e^{-x}}$$
 then f is
A) a bijection
B) an injection only
C) surjection only
D) neither injection nor surjection
5. Let the function $f: R \rightarrow R$ defined by
 $f(x) = 2x + \sin x$ then ' f ' is
a) one-one and onto b) one-one but not onto
c) onto but not one-one
d) neither one-one nor onto
6. Let $f: R \rightarrow R$ be any function. Define
 $g: R \rightarrow R$ by $g(x) = |f(x)|$ for all ' x ' then ' g ' is
A) onto if ' f ' is onto
B) one-one if ' f ' is one-one
C) continuous if ' f ' is continuous
D) neither one - one nor onto
7. Let $f: R \rightarrow R$ and $g: R \rightarrow R$ be two one-one and onto functions such that they are the mirror images of each other about the line $y=a$. If $h(x) = f(x) + g(x)$ then $h(x)$ is
a) one-one onto b) one-one into
c) many-one into d) many-one onto
8. Let $f: R \rightarrow Q$ be a continuous function such that $f(2)=3$ then $f(x)$ is always
A) an even function B) an odd function
C) neither even nor odd
D) an increasing function
9. $f(x) = \ln(x + \sqrt{1+x^2})$ is
A) even function B) odd function
C) Neither even nor odd
D) Constant function
10. A function is matched below against an interval where it is supposed to be increasing (or non decreasing). The incorrect match of the following is

Interval	Function
a) $(-\infty, \infty)$	$x^3 - 3x^2 + 3x + 3$
b) $[2, \infty)$	$2x^3 - 3x^2 - 12x + 6$
c) $(-\infty, \frac{1}{3}]$	$3x^2 - 2x + 1$
d) $(-\infty, -4)$	$x^3 + 6x^2 + 6$

a) a b) b c) c d) d
11. If $f(x)$ is a function that is odd and even simultaneously then $f(3) - f(2)$ is
a) 1 b) -1 c) 0 d) 2
12. The entire graph of $y = x^2 + kx - x + 9$ is strictly above the X-axis if and only if
[IIT 1979]
a) $K < 7$ b) $-5 < K < 7$
c) $K > -5$ d) $K > 7$
13. $f(x) = \frac{\cos x}{\left[\frac{x}{\pi}\right] + \frac{1}{2}}$ (where x is not an integral multiple of π and $[\cdot]$ is G.I.F) is
a) even function b) odd function
c) Neither even nor odd d) cannot decide
14. Let $f: [-10, 10] \rightarrow R$ where
$$f(x) = \sin x + \left[\frac{x^2}{a}\right]$$
 where $[\cdot]$ is G.I.F be an

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odd function then $a \in$

- a) $(-10, 10) - \{0\}$ b) $(0, 10)$
c) $[100, \infty)$ d) $(100, \infty)$

15. The function $f(x) = \int_0^x \log_e \left(\frac{1-x}{1+x} \right) dx$ is

- A) an even function B) an odd function
C) a periodic function D) Neither even nor odd

16. $f: R - \{n\} \rightarrow R$ a function defined by

$$f(x) = \frac{x-m}{x-n} \text{ where } (m \neq n) \text{ f is}$$

- A) one-one & onto B) one-one & into
C) many-one & onto D) many-one & into

17. The domain of $f(x) = \frac{1}{x} + 2^{\sin^{-1}x} + \frac{1}{\sqrt{x-2}}$ is

- A) $(0, \infty)$ B) $(-\infty, 0)$ C) $(1, 3)$ D) $\emptyset$

18. The domain of $f(x) = \frac{\log_2(x+3)}{x^2+3x+2}$ is

- A) $(3, \infty)$ B) $(-3, \infty)$
C) $(1, \infty)$ D) $(-3, \infty) - \{-1, -2\}$

19. The domain of $f(x) = \sqrt{\ln_{(|x|-1)}(x^2+4x+4)}$ is

- A) $[-3, -1] \cup [1, 2]$
B) $(-2, -1) \cup [2, \infty)$
C) $(-\infty, -3] \cup (-2, -1) \cup (2, \infty)$
D) $(-\infty, \infty)$

20. If the function $f(x)$ is defined for $x \in [0, 1]$, then the function $f(2x+3)$ is defined for

- A) $(-\infty, \infty)$ B) $\left[\frac{-3}{2}, -1 \right]$
C) $\left[\frac{-3}{2}, +1 \right]$ D) $(0, \infty)$

21. The domain of $f(x) = \sqrt{\sin^{-1}(\log_2 x)}$ is

- A) $[1, 2]$ B) $[-1, 1]$ C) $[0, 1]$ D) $(1, 2)$

22. The domain of $f(x) = \cos^{-1}([x])$ (where $[.]$ is G.I.F) is

- A) $[-1, 1]$ B) $[0, 1]$ C) $[-1, 2]$ D) $[-1, 2)$

23. The domain of $f(x) = \sqrt{\frac{\log_{0.3}|x-2|}{|x|}}$ is

- A) $[1, 2]$ B) $[2, 3]$ C) $[1, 2) \cup (2, 3]$ D) $[0, 3]$

24. The domain of

$$f(x) = \log_{10} \log_{10} \log_{10} \dots \log_{10} x$$

(‘log’ n times) is

A) $\left(10^{10^{10^{-(n-2)\text{times}}}}, \infty \right)$ B) $(10^{n-2}, \infty)$

C) $\left(10^{10^{10^{-(n-1)\text{times}}}}, \infty \right)$ D) $\left(10^{10^{10^{-(n-3)}}, \infty \right)$

25. The domain of $f(x) = \sqrt{\sin^{-1}(2x) + \frac{\pi}{6}}$ is

- A) $\left[\frac{-1}{4}, \frac{1}{2} \right]$ B) $\left[\frac{-1}{2}, \frac{1}{2} \right]$ C) $\left[\frac{-1}{2}, \frac{1}{4} \right]$ D) $\left[\frac{-1}{4}, \frac{1}{4} \right]$

26. The domain of the function

$$f(x) = \sin^{-1}\left(\frac{4}{3+2\cos x}\right) \text{ is}$$

A) $\left[2n\pi - \frac{\pi}{3}, 2n\pi + \frac{\pi}{3} \right]$ B) $\left[2n\pi - \frac{\pi}{6}, 2n\pi + \frac{\pi}{6} \right]$

C) $\left[2n\pi - \frac{\pi}{2}, 2n\pi + \frac{\pi}{2} \right]$ D) $\left[2n\pi - \frac{\pi}{3}, 2n\pi + \frac{\pi}{2} \right]$

27. The function

$$f(x) = \cot^{-1}\left(\sqrt{x(x+3)}\right) + \cos^{-1}\left(\sqrt{x^2+3x+1}\right)$$

is defined on the set S, where S is

- A) $[-3, 0]$ B) $\{-3, 0\}$ C) $[0, 3]$ D) $\emptyset$

28. The domain of $f(x) = \sin^{-1}[2-4x^2]$ (where $[.]$ is G.I.F) is

A) $\left[\frac{-\sqrt{3}}{2}, \frac{\sqrt{3}}{2} \right]$ B) $\left[\frac{-3}{2}, \frac{3}{2} \right] - \{0\}$

$$C) \left[\frac{-\sqrt{3}}{2}, 0 \right) \cup \left(0, \frac{\sqrt{3}}{2} \right] \quad D) \phi$$

29. If the domain of $f(x)$ is $(0,1)$, then the domain of $f(e^x) + f(\ln|x|)$ is

$$A) (-1, e) \quad B) (1, e) \\ C) (-e, -1) \quad D) (-e, 1)$$

30. The domain of

$$f(x) = \sqrt{\cos(\sin x)} + \sqrt{\log_x \{x\}} \quad (\text{where } \{.\} \text{ is fractional part of } x) \text{ is}$$

$$A) [1, \pi) \quad B) (0, 2\pi) - [1, \pi) \\ C) \left(0, \frac{\pi}{2} \right) - \{1\} \quad D) (0, 1)$$

31. The domain of

$$\frac{[\cos^{-1}(x^4)] + |x - 2 \tan^{-1} x| + \sqrt{\sin(\ln x)}}{\{3x^2 - 7\} + a^{\sqrt{\sin x + 3 \cos x}} + \ln \cos \left(\frac{1}{\sqrt{-x^2}} \right)}$$

(where $[.]$ is G.I.F and $\{.\}$ is fractional part function) is

$$A) (-2, \sqrt{2}) \quad B) (0, 1) \\ C) (-1, 1) \quad D) \phi$$

32. The domain of $f(x) = \sin^{-1}([2 - 3x^2])$ (where $[.]$ is G.I.F) is

$$a) [-1, 1] \quad b) (0, 1) \quad c) [-1, 1] - \{0\} \quad d) \phi$$

33. The domain of $f(x) = \sin^{-1}\left(\frac{1+x^2}{2x}\right)$ is

$$a) (-1, 1) \quad b) [-1, 0] \quad c) \{-1, 1\} \quad d) \{-1, 0\}$$

34. The domain of $f(x) = \cos^{-1}(\sec(\cos^{-1} x)) + \sin^{-1}(\operatorname{cosec}(\sin^{-1} x))$ is

$$a) (-1, 1) \quad b) \{-1, 1\} \quad c) [-1, 1] \quad d) \phi$$

35. The range of $\tan(\log x)$ is

$$A) (0, \infty) \quad B) (1, \infty) \quad C) (e, \infty) \quad D) (-\infty, \infty)$$

36. The range of $f(x) = \frac{e^{-x}}{1+[x]}$ is (where $[.]$

is G.I.F)

$$A) \mathbb{R} \quad B) \mathbb{R} - \{0\} \\ C) \mathbb{R} - [-1, 0) \quad D) [0, \infty)$$

37. The range of $f(x) = \frac{1}{|\sin x|} + \frac{1}{|\cos x|}$ is

$$A) [2\sqrt{2}, \infty) \quad B) (\sqrt{2}, 2\sqrt{2}) \\ C) (0, 2\sqrt{2}) \quad D) (2\sqrt{2}, 4)$$

38. The range of $f(x) = \sin^2 x - 5 \sin x - 6$ is

$$A) [-10, 0] \quad B) [-1, 1] \\ C) [0, \pi] \quad D) \left[\frac{-49}{4}, 0 \right]$$

39. The range of $f(x) = \frac{x - [x]}{1 - [x] + x}$

(where $[.]$ is G.I.F) is

$$A) \left[0, \frac{1}{2} \right] \quad B) [0, 1] \quad C) (0, \frac{1}{2}] \quad D) [0, \frac{1}{2})$$

40. If $f: \mathbb{R} \rightarrow S$ defined by

$$f(x) = \sin x - \sqrt{3} \cos x + 1 \text{ is an onto function, then } S =$$

$$A) [1, 3] \quad B) [-1, 3] \quad C) [0, 1] \quad D) [-1, 1]$$

41. Let $A = \left\{ x / 0 \leq x < \frac{\pi}{2} \right\}$ and $f: \mathbb{R} \rightarrow A$ is an onto function given by $f(x) = \tan^{-1}(x^2 + x + \lambda)$ where

$$A) \lambda > 0 \quad B) \lambda \leq \frac{1}{4} \quad C) \lambda = \frac{1}{4} \quad D) \lambda \geq \frac{1}{8}$$

42. The range of $f(x) = (7-x)_{P_{(x-3)}}$ is

$$A) \{1, 2, 3, 4, 5\} \quad B) \{3, 4, 5\} \\ C) \{1, 2, 3\} \quad D) \{1, 2, 3, 4\}$$

43. The range of $f(x) = \frac{x^2 - x + 1}{x^2 + x + 1}$ is

$$A) \left[\frac{1}{3}, 3 \right] \quad B) \left[\frac{1}{2}, 2 \right] \\ C) [0, 1] \quad D) [-1, 1]$$

44. The range of $f(x) = \left[\frac{1}{\sin\{x\}} \right]$ is (where $\{.\}$ is fractional part and $[.]$ is G.I.F)
A) $\{1, -1\}$ B) $\{0\}$ C) $\mathbb{N}$ D) $\mathbb{Z}$
45. The range of $f(x) = \frac{\tan(\pi[x^2 - x])}{1 + \sin(\cos x)}$ (where $[.]$ is G.I.F) is
A) $\left(0, \frac{\pi}{2}\right)$ B) $\{0, 1\}$
C) $\{0\}$ D) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
46. The range of $f(x) = \sqrt{a-x} + \sqrt{x-b}$ is (where $a > b > 0$)
A) $[\sqrt{a-b}, \sqrt{2(a-b)}]$
B) $[\sqrt{a-b}, \sqrt{(a+b)}]$
C) $[a, b]$ D) (a, b)
47. If $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^3 + 3x^2 + 10x + 2 \sin x$ then the range of the function is given by
A) $(-\infty, \infty)$ B) $[0, \infty)$
C) $(-\infty, 0]$ D) $\emptyset$
48. If $f(x) = x^2 + 2bx + 2c^2$ and $g(x) = -x^2 - 2cx + b^2$ such that minimum of $f(x) >$ maximum of $g(x)$ then [IIT 2003]
a) $|c| < |b| \sqrt{2}$ b) $|c| > |b| \sqrt{2}$
c) $|c| < |b|$ d) None
49. The range of $f(x) = \cot^{-1}(2x - x^2)$ is
a) $(0, \pi)$ b) $\left(\frac{\pi}{4}, \pi\right)$ c) $\left(\frac{\pi}{4}, \pi\right]$ d) $\left[\frac{\pi}{4}, \pi\right)$
50. The sum of the maximum and minimum values of $f(x) = \sin^{-1}(2x) + \cos^{-1}(2x) + \sec^{-1}(2x)$ is
a) π b) $\frac{\pi}{2}$ c) 2π d) $\frac{3\pi}{2}$
51. If a, b, c, d, e are +ve real numbers such that $a + b + c + d + e = 8$ and $a^2 + b^2 + c^2 + d^2 + e^2 = 16$ then the range of 'e' is
- a) $\left[0, \frac{16}{5}\right]$ b) $\left[1, \frac{5}{3}\right]$ c) $\left(0, \frac{5}{16}\right]$ d) $[-1, 1]$
52. Domain and range of $f(x) = \sin^{-1}(\cos^{-1}[x])$, where $[.]$ is G.I.F, are
A) $[1, 2), \{0\}$ B) $[0, 1], \{-1, 0, 1\}$
C) $[-1, 1], \left\{0, \sin^{-1}\left(\frac{\pi}{2}\right), \sin^{-1}(\pi)\right\}$
D) $[-1, 1], \left\{-\frac{\pi}{2}, 0, \frac{\pi}{2}\right\}$
53. If $f(x) = \log_{[x-1]}\left(\frac{|x|}{x}\right)$, where $[.]$ is G.I.F then domain and range are
A) $(2, \infty), (0, 1)$ B) $[3, \infty), \{0\}$
C) $[3, \infty), \{0, 1\}$ D) $(-\infty, \infty); \{0\}$
54. Let $f(x)$ and $g(x)$ be bijective functions where $f: \{a, b, c, d\} \rightarrow \{1, 2, 3, 4\}$ and $g: \{3, 4, 5, 6\} \rightarrow \{w, x, y, z\}$ respectively. The number of elements in the range set of $g(f(x))$ is
A) 1 B) 2 C) 3 D) 4
55. If $f(x) = x^3 + 3x^2 + 4x + a \sin x + b \cos x$ $\forall x \in \mathbb{R}$ is an injection then the greatest value of $a^2 + b^2$ is
A) 1 B) 2 C) $\sqrt{2}$ D) $2\sqrt{2}$
56. The domain and range of $f(x) = \sin \left\{ \log \left(\frac{\sqrt{4-x^2}}{1-x} \right) \right\}$ are
A) $(-2, 1)$ and $[-1, 1]$ B) $(1, 3)$ and $[-1, 1]$
C) $[-2, 1]$ and $[-1, 1]$ D) $(0, \infty)$ and $[-1, 1]$
57. If 'f' and 'g' are two functions such that composite function $g \circ f$ is one-one then
A) 'f' must be one-one
B) 'g' must be one-one
C) both are one one
D) both 'f' and 'g' are onto

58. If $g(x) = 1 + x - [x]$ and

$$f(x) = \begin{cases} -1 & \text{if } x < 0 \\ 0 & \text{if } x = 0 \\ 1 & \text{if } x > 1 \end{cases} \text{ then } f \circ g(x) =$$

- a) $-1 \forall x \in R$ b) $0 \forall x \in R$
c) $1 \forall x \in R$ d) $5 \forall x \in R$

59. If $f(x) = \begin{cases} 1+x; & \text{if } 0 \leq x \leq 2 \\ 3-x; & \text{if } 2 < x \leq 3 \end{cases}$ then

$$(f \circ f)(x) =$$

- a) $1-x; 0 \leq x \leq 2$ b) $2-x; 2 < x \leq 3$
c) $3+x; 2 < x \leq 3$ d) $4-x; 2 < x \leq 3$
 $2+x; 0 \leq x \leq 1$
c) $2-x; 1 < x \leq 2$ d) does not exist
 $4-x; 2 < x \leq 3$

60. If for $x > 0$, $f(x) = (a - x^n)^{\frac{1}{n}}$;

$g(x) = x^2 + px + q$; $p, q \in R$ and the equation $g(x) - x = 0$ has imaginary roots then the number of real roots of the equation

$$g(g(x)) - f(f(x)) = 0 \text{ is}$$

- a) 0 b) 2 c) 4 d) n

61. If $g(x) = 1 + \sqrt{x}$ and

$$f(g(x)) = 3 + 2\sqrt{x} + x \text{ then } f(x) =$$

- a) $1 + 2x^2$ b) $2 + x^2$ c) $1 + x$ d) $2 + x$

62. If $f(x) = \sin x + \cos x$, $g(x) = x^2 - 1$ then $g(f(x))$ is invertible in the domain

[IIT 2004]

- a) $\left[0, \frac{\pi}{2}\right]$ b) $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$ c) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ d) $[0, \pi]$

63. If 'f' is a decreasing odd function, then f^{-1} is

- A) odd and decreasing B) odd and increasing
C) even and decreasing D) even and increasing

64. Let $f(x) = (x+1)^2 - 1$; $x \geq -1$ then the set $\{x : f(x) = f^{-1}(x)\}$ is [IIT 1995]

a) $\left\{0, -1, \frac{-3+i\sqrt{3}}{2}, \frac{-3-i\sqrt{3}}{2}\right\}$ b) $\{0, 1, -1\}$

c) $\{0, -1\}$ d) $\emptyset$

65. If the function $f(x) = \sin x + \cos(ax)$ is periodic, then 'a' is

- a) any real number b) any integer
c) any rational number d) no such 'a'

66. The period of

$$\frac{|\sin(4x)| + |\cos(4x)|}{|\sin(4x) - \cos(4x)| + |\sin(4x) + \cos(4x)|} \text{ is}$$

- a) $\frac{\pi}{4}$ b) $\frac{\pi}{2}$ c) $\frac{\pi}{8}$ d) π

67. If $f(x) = \cos x + \{x\}$ where $\{.\}$ is fractional part function then the period of $f(x)$ is

- a) 2π b) 1
c) $\frac{\pi}{2}$ d) Does not exist

68. The period of

$$f(x) = \sin x + \tan\left(\frac{x}{2}\right) + \sin\left(\frac{x}{2^2}\right) + \tan\left(\frac{x}{2^3}\right) + \dots + \sin\left(\frac{x}{2^{n-1}}\right) + \tan\left(\frac{x}{2^n}\right)$$

- a) 2π b) $2^n \pi$ c) $2^n \frac{\pi}{3}$ d) $3^n \pi$

69. Period of $f(x) = \sin((\cos x) + x)$ is

- a) Does not exist b) π
c) $\frac{\pi}{2}$ d) 2π

70. The period of

$$f(x) = e^{x - [x] + |\cos(\pi x)| + |\cos(2\pi x)| + \dots + |\cos(n\pi x)|}$$

(where $[.]$ is G.I.F) is

- a) 1 b) $\frac{1}{2}$ c) $\frac{1}{4}$ d) $1/5$

71. The period of the function

$$|\sin^3\left(\frac{x}{2}\right)| + |\cos^5\left(\frac{x}{5}\right)| \text{ is}$$

- a) 2π b) 10π c) 8π d) 5π

72. If 'f' is periodic, 'g' is non-periodic,

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$f(g(x))$ is periodic and $f(2) = 3, g(4) = 7$ then $g(6)$ is

- a) 13 b) 15 c) 11 d) 0

73. Let $f(x)$ be a real valued function with domain R such that $f(x+p)$

$$= 1 + \left[2 - 3f(x) + 3(f(x))^2 - (f(x))^3 \right]^{1/3}$$

holds good $\forall x \in R$ and for some +ve constant 'p' then the period of $f(x)$ is

- a) $\frac{p}{2}$ b) p c) $2p$ d) $\frac{p}{3}$

74. If $f(x)$ and $g(x)$ are periodic function and non-periodic function respectively then $f(g(x))$ is

- a) always periodic b) never periodic
c) periodic when $g(x)$ is a linear function of 'x'
d) can not say

75. If $f(a-x) = f(a+x)$ and

$f(b-x) = f(b+x) \forall x \in R$ where $a, b(a > b)$ are constants then the period of $f(x)$ is

- a) $2a$ b) $2b$ c) $3a$ d) b

76. The period of $f(x) = [x] + [2x] + [3x] + [4x]$

$$+ \dots \dots \dots [nx] - \frac{n(n+1)}{2}x \text{ (where } n \in N \text{) is}$$

- a) n b) 1 c) $1/n$ d) 5

77. Consider the real valued function

satisfying $2f(\sin x) + f(\cos x) = x$, then

$$f\left(\frac{1}{2}\right) =$$

- A) 1 B) 2 C) 0 D) 4

78. Let $f(x)$ be defined for all $x > 0$ and be continuous. Let $f(x)$ satisfy the relation

$$f\left(\frac{x}{y}\right) = f(x) - f(y) \text{ for all 'x' and 'y' and}$$

$f(e) = 1$ then [IIT - 1995]

$$f\left(\frac{1}{x}\right) \rightarrow 0 \text{ as } x \rightarrow 0$$

$$x f(x) \rightarrow 0 \text{ as } x \rightarrow 0 \quad D) f(x) = \ln x$$

79. If $f: R \rightarrow R$ such that $f(x - f(y)) = f(f(y)) + x f(y) + f(x) - 1 \forall x, y \in R$ then $f(x)$ is

- a) $1 + \frac{x^2}{4}$ b) $1 - \frac{x^2}{2}$ c) $1 + \frac{x^2}{2}$ d) $1 - \frac{x^2}{4}$

80. A function 'f' well defined $\forall x, y \in R$ is such that $f(1) = 2, f(2) = 8$ and

$f(x+y) - kxy = f(x) + 2y^2$, where 'k' is some constant then $f(x)$ is

- a) x^2 b) $3x^2$ c) $2x^2$ d) $4x^2$

81. If $f(x+y) = f(x)f(y) \forall x, y \in R$ and

$$f(0) \neq 0 \text{ then } F(x) = \frac{f(x)}{1 + (f(x))^2} \text{ is}$$

- a) an even function b) an odd function
c) both d) Neither even nor odd function

82. If 'f' is a polynomial function satisfying

$$2 + f(x)f(y) = f(x) + f(y) + f(xy)$$

$\forall x, y \in R$ and if $f(2) = 5$ then the value of

$$f(f(2)) =$$

- a) 25 b) 16 c) 26 d) 14

83. Let $g(x) = f(x) - 1$.

If $f(x) + f(1-x) = 2 \forall x \in R$ then $g(x)$ is symmetrical about

- a) the origin b) the line $x = \frac{1}{2}$
c) the point $(1, 0)$ d) the point $\left(\frac{1}{2}, 0\right)$

84. For $x \in R$, the function $f(x)$ satisfies

$$2f(x) + f(1-x) = x^2 \text{ then the value of } f(4) \text{ is}$$

a) $\frac{13}{3}$ b) $\frac{43}{3}$ c) $\frac{23}{3}$ d) $\frac{23}{4}$

85. If f is a function such that $f(0) = 2; f(1) = 3$ and $f(x+2) = 2f(x) - f(x+1) \forall x \in R$ then $f(5) =$

a) 7 b) 13 c) 1 d) 5

86. If $[x]$ and $\{x\}$ represent the integral and fractional parts of x respectively, then the

value of $\sum_{r=1}^{2000} \frac{\{x+r\}}{2000}$ is

A) x B) $[x]$ C) $\{x\}$ D) $x + 2001$

87. Let $A = \{1, 2, 3, 4, 5\}$. If ' f ' is a bijective function from A to A , then the number of such functions for which $f(k) \neq k$, $k=1, 2, 3, 4, 5$ is

A) 5^5 B) 120 C) 44 D) $5^5 - 120$

88. Total number of solutions of

$2^x + 3^x + 4^x - 5^x = 0$ is

A) 0 B) 1
C) 2 D) infinitely many

89. The set of values of ' a ' for which the function $f: R \rightarrow R$ given by

$f(x) = x^3 + (a+2)x^2 + 3ax + 5$ is one-one is

A) $[-2, 4]$ B) $(1, 3)$ C) $(1, 4)$ D) $(1, 5)$

90. A function $f: R \rightarrow R$ satisfies

$\sin x \cos y [f(2x+2y) - f(2x-2y)]$
 $= \cos x \sin y [f(2x+2y) + f(2x-2y)]$.

If $f^1(0) = \frac{1}{2}$ then $4f^{11}(x) + f(x)$ is

a) -1 b) 2 c) 4 d) 0

91. If $f(x) = 2x + |x|$, $g(x) = \frac{1}{3}(2x - |x|)$ and $h(x) = f(g(x))$ then the domain of $\sin^{-1}(h(h(h \dots h(x))))$ is (h being repeated n times)

a) $[-1, 1]$ b) $(0, 1)$ c) $(-1, 1)$ d) $\left[\frac{1}{2}, 1\right]$

92. If for a real number x , then the value of

$\left[\frac{n+1}{2}\right] + \left[\frac{n+2}{4}\right] + \left[\frac{n+4}{8}\right] + \left[\frac{n+8}{16}\right] + \dots =$

($[x]$ denotes G.I.F, $n \in N$)

a) n b) $n-1$ c) $n+1$ d) $n+2$

93. Let R be the set of real numbers and $f: R \rightarrow R$ be such that for all $x, y \in R$, $|f(x) - f(y)| \leq |x - y|^3$ then $f(x)$ is [IIT 1988]

a) a quadratic function b) constant function
c) a cubic function d) linear function

MULTIANSWER QUESTIONS

94. $f: [0, \infty) \rightarrow R$ given by $f(x) = \frac{x}{x+1}$ is

a) one-one b) onto
c) into d) many-one

95. $f: (e, \infty) \rightarrow R$ defined by

$f(x) = \log(\log(\log x))$ then

a) many-one b) into
c) one-one d) onto

96. $f(x) = \begin{cases} x & \text{if } x \text{ is rational} \\ 0 & \text{if } x \text{ is irrational} \end{cases}$ and

$g(x) = \begin{cases} 0 & \text{if } x \text{ is rational} \\ x & \text{if } x \text{ is irrational} \end{cases}$ then $f - g$ is

[IIT-2005]

(A) one-one and into
(B) neither one-one nor onto
(C) many one and onto
(D) one-one and onto

97. If $f(x) = \cos[\pi^2]x + \cos[-\pi^2]x$ (where $[x]$ stands for the greatest integer function) then [IIT-1991]

(A) $f(\pi/2) = -1$ (B) $f(\pi) = 1$
(C) $f(-\pi) = 0$ (D) $f(\pi/4) = 2$

98. The $f: R \rightarrow R$ be any function. Define $g: R \rightarrow R$ by $g(x) = |f(x)|$ for all x then g is [IIT - 2000]

(A) onto if f is onto
(B) one-one if f is one-one
(C) continuous if f is continuous

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- (D) differentiable if f is differentiable
99. Let $f(x) = [x]^2 + [x+1] - 3$ where $[x] \leq x$ then and $f: \mathbb{R} \rightarrow \mathbb{R}$
- a) many-one b) one-one
c) onto d) into
100. The domain of $f(x) = \frac{1}{\sqrt{\log_{1/2}(x^2 - 7x + 13)}}$ is
- a) (3,4) b) [3,4]
c) $(-\infty, 3] \cup [1, \infty)$ d) (1,3)
101. The domain of definition of $f(x) = \frac{\log_2(x+3)}{x^2 + 3x + 2}$ is [IIT - 2001]
- (A) $\mathbb{R} - \{-1, -2\}$ (B) $(-2, \infty)$
(C) $\mathbb{R} - \{-1, -2, -3\}$ (D) $(-3, \infty) - \{-1, -2\}$
102. $f(x) = \log_{(100x)}\left(\frac{2\log_{10} x + 1}{-x}\right)$ exists if $x \in$
- a) $(0, 10^{-2})$ b) $\left(10^{-2}, 10^{-\frac{1}{2}}\right)$
c) $(10^2, 10^3)$ d) $(10^8, \infty)$
103. If $f(x)$ is defined on the domain $[0,1]$ then $f(2 \sin x)$ is defined on
- a) $\bigcup_{n \in \mathbb{I}} \left[2n\pi, 2n\pi + \frac{\pi}{6}\right]$ b) $\bigcup_{n \in \mathbb{I}} \left[2n\pi + \frac{5\pi}{6}, (2n+1)\pi\right]$
c) $\bigcup_{n \in \mathbb{I}} \left[2n\pi, 2n\pi + \frac{\pi}{2}\right]$ d) $\bigcup_{n \in \mathbb{I}} \left[2n\pi + \frac{\pi}{6}, 2n\pi + \frac{5\pi}{6}\right]$
104. For the function $f(x) = \log_{10}(3x^2 - 4x + 5)$
- a) Domain is $(0, \infty)$ b) range is $\mathbb{R}$
c) Domain is $\mathbb{R}$ d) range is $\left[\log_{10}\left(\frac{11}{3}\right), \infty\right)$
105. For the function $f(x) = \cos^{-1} \sqrt{\log_{[x]}\left(\frac{|x|}{x}\right)}$ (where $[.]$ G.I.F) is
- a) Domain is $[1, \infty)$ b) Domain is $[2, \infty)$
c) range is $[0, \infty)$ d) range is $\left\{\frac{\pi}{2}\right\}$
106. Range of the function $f(x) = \frac{x^2 + x + 2}{x^2 + x + 1}; x \in \mathbb{R}$ is [IIT - 2003]

- (A) $(1, \infty)$ (B) $(1, 11/7]$
(C) $(1, 7/3]$ (D) $[1, 7/5]$
107. If the functions $f(x) = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$ and $g(x) = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ are identical functions then their
- a) Domain is $[-1,1]$ b) Domain is $[0,1]$
c) range is $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ d) range is $\left[0, \frac{\pi}{2}\right]$
108. Let $f: X \rightarrow Y, f(x) = \sin x + \cos x + 2\sqrt{2}$ is invertible then
- a) $X = \left[\frac{\pi}{4}, \frac{5\pi}{4}\right]$ b) $X = \left[\frac{\pi}{4}, \frac{\pi}{2}\right]$
c) $Y = [-1,1]$ d) $Y = [\sqrt{2}, 3\sqrt{2}]$
109. If $f(x) = 3x - 5$ then $f^{-1}(x)$ [IIT - 1998]
- (A) is given by $\frac{1}{3x-5}$ (B) is given by $\frac{x+5}{3}$
(C) does not exist because f is not one-one
(D) does not exist because f is not onto
110. Let $f(x) = \frac{\alpha x}{x+1}, x \neq -1$ then the value of α so that $f(f(x)) = x$ is [IIT - 2001]
- (A) $\sqrt{2}$ (B) $-\sqrt{2}$ (C) 1 (D) -1
111. Let $f(x) = \sin x + \cos x$ and $g(x) = x^2 - 1$ then domain for which $g \circ f$ is invertible is [IIT - 2004]
- (A) $\left(0, \frac{\pi}{2}\right)$ (B) $\left(\frac{\pi}{2}, \frac{2\pi}{3}\right)$
(C) $\left(-\frac{\pi}{2}, \frac{\pi}{3}\right)$ (D) $\left(-\frac{\pi}{4}, \frac{\pi}{4}\right)$
112. If X and Y are two non-empty sets where $f: X \rightarrow Y$ is function is defined such that $f(C) = \{f(x) : x \in C\}$ for $C \subseteq X$ and $f^{-1}(D) = \{x : f(x) \in D\}$ for $D \subseteq Y$ and for any $A \subseteq Y, B \subseteq Y$ then [IIT - 2005]
- (A) $f^{-1}(f(A)) = A$

(B) $f^{-1}(f(A)) = A$ only if $f(X) = Y$

(C) $f(f^{-1}(B)) = B$ only if $B \subseteq f(X)$

(D) $f(f^{-1}(B)) = B$

113. All periodic functions are

- a) one-one b) many-one
c) invertible
d) invertible by restricting the domain

114. Let 'n' be +ve integer with

$$f(n) = \lfloor 1 \rfloor + \lfloor 2 \rfloor + \lfloor 3 \rfloor + \dots + \lfloor n \rfloor \quad \text{and}$$

$p(x), Q(x)$ be polynomials in 'x' such

that $f(n+2) = p(n) \cdot f(n+1) + Q(n) \cdot f(n)$

for all $n \geq 1$, then

- a) $p(x) = x+3$ b) $Q(x) = -x-2$
c) $p(x) = -x-2$ d) $Q(x) = x+3$

115. If $\{x\}$ and $[x]$ denote the fractional and integral parts of 'x' and

$\{x+1\} + 2x = 4[x+1] - 6$ then x is

- a) 1 b) $\frac{8}{3}$ c) 0 d) -1

$$116. f(x) = \begin{cases} 0 & \text{for } x = 0 \\ x^2 \sin\left(\frac{\pi}{x}\right) & \text{for } -1 < x < 1, x \neq 0 \\ x|x| & \text{for } x \geq 1 \text{ or } x \leq -1 \end{cases}$$

then $f(x)$ is

- a) an odd function b) an even function
c) neither even nor odd
d) $f^{-1}(x)$ is an even function

117. The domain of definition of the function $f(x)$ given by the equation $2^x + 2^y = 2$ is

[IIT - 2000]

- (A) $0 < x \leq 1$ (B) $0 \leq x \leq 1$
(C) $-\infty < x \leq 0$ (D) $-\infty < x < 1$

COMPREHENSION QUESTIONS

(P) $f: A \rightarrow B$ is said to be injective if distinct elements in A have distinct images in B and surjective if $f(A) = B$. Now answer the following.

118. If the function $f: A \rightarrow B$ defined by

$$f(x) = \sqrt{\frac{1 + \cos(2x)}{2}} \text{ is injective then A can be}$$

- a) $[0, \pi]$ b) $[-\pi, \pi]$
c) $\left[-\frac{\pi}{2}, 0\right]$ d) $[-\pi, 0]$

119. If the function $f: R \rightarrow B$ defined by

$$f(x) = \sqrt{x^2} \text{ is surjective then B is}$$

- a) $(-\infty, 0]$ b) $[0, \infty)$ c) $(0, \infty)$ d) R

120. The functions $f: R \rightarrow B$ defined by

$$f(x) = [x] + [-x] \text{ (where } [.] \text{ is G.I.F.) is surjective then B=}$$

- a) R b) $[0, 1]$ c) $[-1, 0]$ d) $\{-1, 0\}$

(P) $f: [2, \infty) \rightarrow [1, \infty)$ defined $f(x) = 2^{x^4 - 4x^2}$

$$g: \left[\frac{\pi}{2}, \pi\right] \rightarrow A \text{ defined by } g(x) = \frac{\sin x + 4}{\sin x - 2}$$

be two invertible functions then

121. $f^{-1}(x) =$

- a) $\sqrt{2 + \sqrt{4 - \log_2 x}}$ b) $\sqrt{2 + \sqrt{4 + \log_2 x}}$
c) $\sqrt{2 - \sqrt{4 + \log_2 x}}$ d) $2 + \sqrt{4 + \log_2 x}$

122. $A =$

- a) $[-5, -2]$ b) $[2, 5]$ c) $[-5, 2]$ d) $[-3, -2]$

123. The domain of $f^{-1}(g^{-1}(x))$ is

- a) $[-5, \sin 1]$ b) $\left[-5, \frac{\sin 1}{2 - \sin 1}\right]$

- c) $\left[-5, \frac{-(4 + \sin 1)}{2 - \sin 1}\right]$ d) $[-1, 1]$

(P) Let $f(x) = \frac{1}{2} \left[f(xy) + f\left(\frac{x}{y}\right) \right] \quad \forall x, y \in R^+$

such that $f(1) = 0, f^{-1}(1) = 2$. Now answer

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the following.

124. $f(x) - f(y) =$

- a) $f\left(\frac{y}{x}\right)$ b) $f\left(\frac{x}{y}\right)$ c) $f(2x)$ d) $f(2y)$

125. $f^{-1}(3) =$

- a) $\frac{1}{3}$ b) $\frac{2}{3}$ c) $\frac{1}{2}$ d) $\frac{1}{4}$

126. $f(e) =$

- a) 2 b) 1 c) 3 d) 4

(P) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ is a function satisfying

$$f(2-x) = f(2+x) \text{ and } f(20-x) = f(x)$$

$\forall x \in \mathbb{R}$. Now answer the following.

127. If $f(0) = 5$ then the minimum possible number of values of 'x' satisfying

$$f(x) = 5 \text{ for } x \in [0, 170] \text{ is}$$

- a) 21 b) 12 c) 11 d) 22

128. The graph of $y = f(x)$ is symmetrical about

- a) $x = 16$ b) $x = 5$ c) $x = 8$ d) $x = 20$

129. If $f(2) \neq f(6)$ then the period of $f(x)$ is

- a) 1 b) may or may not be 1
c) can not be 1 d) Non periodic

(P) Consider two Quadratic polynomials

$$C_a: y = \frac{x^2}{4} - ax + a^2 + a - 2$$

$$C: y = 2 - \frac{x^2}{4}$$

130. If the origin lies between the zeroes of the polynomial C_a then the number of integral values of 'a' is

- a) 1 b) 2 c) 3 d) more than 3

131. If 'a' varies then the equation of the locus of the vertex C_a is

- a) $x - 2y - 4 = 0$ b) $2x - y - 4 = 0$
c) $x - 2y + 4 = 0$ d) $2x + y - 4 = 0$

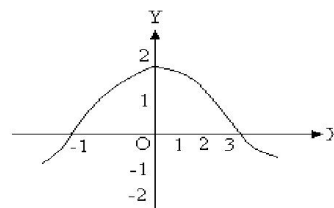
132. For $a=3$ if the lines $y = m_1x + c_1$ and $y = m_2x + c_2$ are common tangents to

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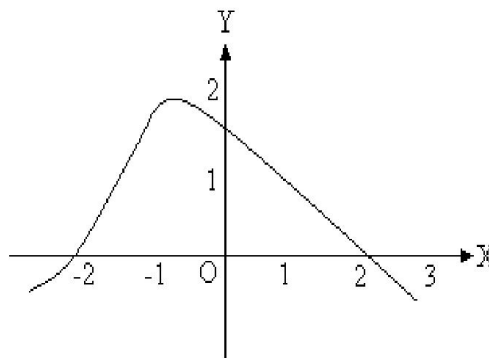
the graph of C_a and C then the value of $m_1 + m_2$ is

- a) -6 b) -3 c) $\frac{1}{2}$ d) $-\frac{1}{2}$

(P) The graph of $y = f(x)$ is given below



Now answer the following.

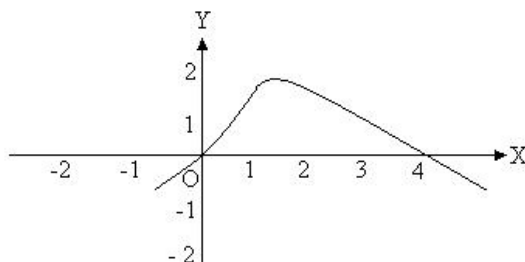


133.

is the graph of

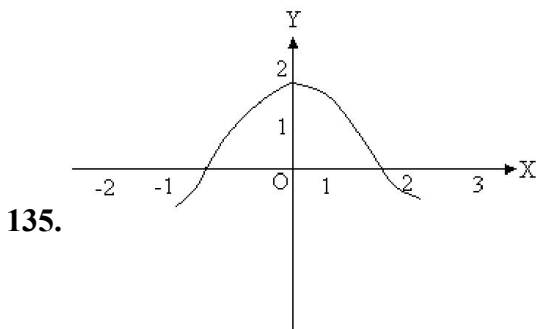
- a) $f(x-1)$ b) $f(x+1)$
c) $f(2x)$ d) $f\left(\frac{x}{2}\right)$

134.



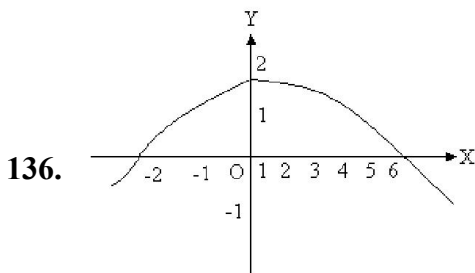
is the graph of

- a) $f\left(x - \frac{1}{2}\right)$ b) $f(x-2)$
c) $f(x+1)$ d) $f(x-1)$



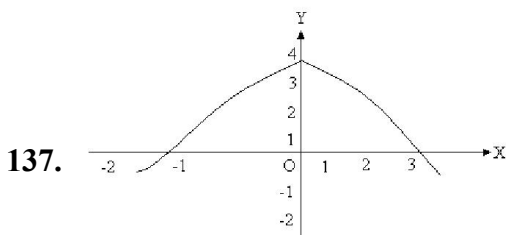
is the graph of

- a) $f(3x)$ b) $f(2x)$
c) $f\left(\frac{x}{2}\right)$ d) $f\left(\frac{x}{3}\right)$



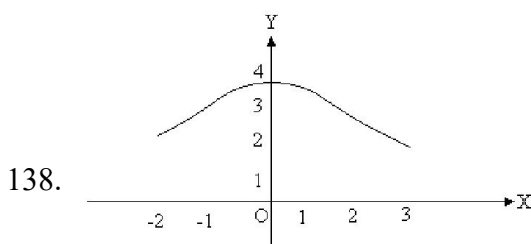
is the graph of

- a) $f\left(\frac{x}{2}\right)$ b) $f\left(\frac{x}{3}\right)$ c) $f(2x)$ d) $f(3x)$



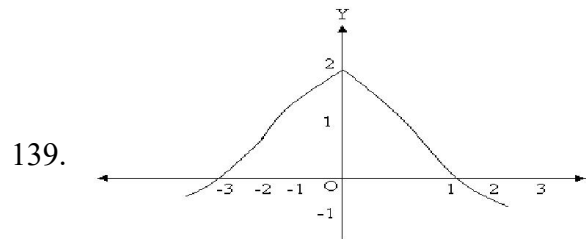
is the graph of

- a) $y = 4f(x)$ b) $y = f\left(\frac{x}{4}\right)$
c) $y = 2f(x)$ d) $y = f(x) + 2$



is the graph of

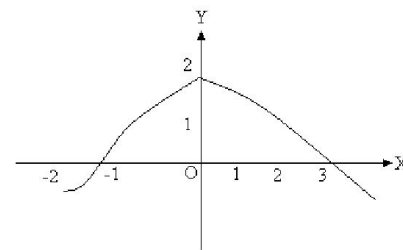
- a) $y = f(x) - 2$ b) $y = f(x) + 2$
c) $y = f(x) + 4$ d) $y = f(x) - 4$



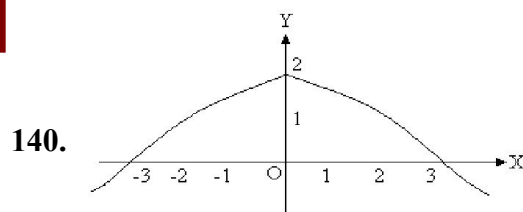
is the graph of

- a) $f(-x)$ b) $f\left(\frac{x}{2}\right)$
c) $(f(x))^{-1}$ d) $\frac{1}{f(x)}$

(P) The graph of $y = f(x)$ is

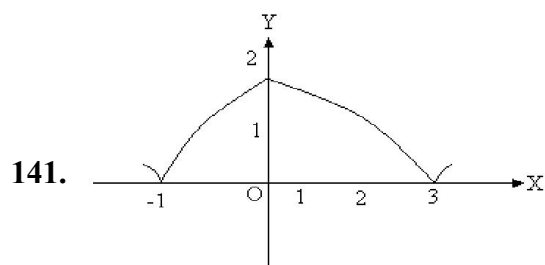


Now answer the following



is the graph of

- a) $y = |f(x)|$ b) $y = |f(|x|)|$
c) $y = f(|x|)$ d) $|y| = f(x)$

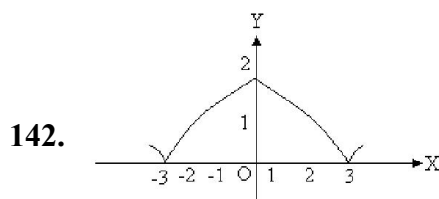


is the graph of

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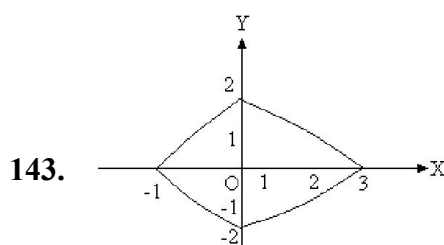
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- a) $y = |f(x)|$ b) $y = |f(|x|)|$
 c) $y = f(|x|)$ d) $|y| = f(x)$



is the graph of

- a) $y = |f(x)|$ b) $y = |f(|x|)|$
 c) $y = f(|x|)$ d) $|y| = f(x)$



is the graph of

- a) $y = |f(x)|$ b) $y = |f(|x|)|$
 c) $y = f(|x|)$ d) $|y| = f(x)$

(P) Let 'f' be a function satisfying

$$f(x) = \frac{a^x}{a^x + \sqrt{a}} = g_a(x) \quad (a > 0)$$

144. Let $f(x) = g_9(x)$ then the value of

$$\left[\sum_{r=1}^{1995} f\left(\frac{r}{1996}\right) \right] = \text{ (where } [.] \text{ is G.I.F)}$$

- a) 995 b) 996 c) 997 d) 998

145. Let $f(x) = g_4(x)$ then $\sum_{r=1}^{1996} f\left(\frac{r}{1997}\right) =$

- a) 100 b) even
 c) odd d) neither even nor odd

146. the value of $g_5(x) + g_5(1-x)$ is

- a) 1 b) 5 c) 10 d) 6

147. The value of $\sum_{r=1}^{2n-1} 2f\left(\frac{r}{2n}\right) =$

- a) 0 b) $2n-1$ c) $2n$ d) $2n-2$

148. If the value of $\sum_{r=0}^{2n} f\left(\frac{r}{2n+1}\right) = \frac{1}{1+\sqrt{a}} + 987$ then the value of 'n' is

- a) 493 b) 494 c) 987 d) 988

MATRIX - MATCHING QUESTIONS

149. Match the following functions with their ranges

COLUMN-I

COLUMN-II

- a) $f(x) = \log_3(5 + 4x - x^2)$ p) function is not defined
 b) $f(x) = \log_3(x^2 - 4x - 5)$ q) $[0, \infty)$
 c) $f(x) = \log_3(x^2 - 4x + 5)$ r) $(-\infty, 2]$
 d) $f(x) = \log_3(4x - 5 - x^2)$ s) $\mathbb{R}$

150. Let $f(x) = \frac{x^2 - 6x + 5}{x^2 - 5x + 6}$

Now match the following [IIT 2007]

COLUMN-I

- a) if $-1 < x < 1$ then $f(x)$ satisfies
 b) if $1 < x < 2$ then $f(x)$ satisfies
 c) if $3 < x < 5$ then $f(x)$ satisfies
 d) if $x > 5$ then $f(x)$ satisfies

COLUMN-II

- p) $0 < f(x) < 1$
 q) $f(x) < 0$
 r) $f(x) > 0$
 s) $f(x) < 1$

ASSERTION - REASON QUESTIONS

The following questions consist of two statements, one labelled as 'Statement-I' and the other 'Statement-II'. You are to examine these two statements carefully and decide if the Statement-I and the Statement-II are individually true and if so, whether the Statement-II is the correct explanation for the given Statement-I. Select your answer to these items using the codes given below and then select the correct option.

- (A) Both S-I and S-II are individually true and R is the correct explanation of A
 (B) Both S-I and S-II are individually true but S-II is not the correct explanation of S-I
 (C) S-I is true but S-II is false

(D) S-I is false but S-II is true

151. Let $f(\theta) = \frac{\sin^2 \theta \cos \theta}{\sin \theta + \cos \theta} - \frac{1}{4} \tan\left(\frac{\pi}{4} - \theta\right)$

$\forall \theta \in R - \left\{n\pi - \frac{\pi}{4}\right\}; n \in Z$ then

S-I: The largest and the smallest values of

$f(\theta)$ differ by $\frac{1}{\sqrt{2}}$

S-II: $a \sin x + b \cos x + c$

$\in [c - \sqrt{a^2 + b^2}, c + \sqrt{a^2 + b^2}] \quad \forall x \in R$

where $a, b, c \in R$

152. S-I: If $f(x)$ is odd function and $g(x)$ is even function then $f(x) + g(x)$ is neither even nor odd

S-II: Odd function is symmetrical in opposite quadrants and even function is symmetrical about the y-axis

153. S-I: If $f(x) = \frac{x+1}{x-1} (x \neq 1)$ then

$(f \circ f \circ f \circ f \circ f)(x) = f(x)$

S-II: $(f \circ f)(x) = f(x)$

154. S-I: If $f(x) = 2x^3 + 7x - 5$ then the value of $f^{-1}(4)$ is 1

S-II: A function $y = f(x)$ is invertible if it is one one onto.

155. S-I: If $f(x) = \frac{a^x}{a^x + \sqrt{a}} (a > 0)$ then

$\sum_{r=1}^{2n-1} 2f\left(\frac{r}{2n}\right) = 2n - 1$

S-II: $f(x) + f(1-x) = 1 \quad \forall x$

INTEGER QUESTIONS

156. If 'f' is a polynomial such that

$f\left(\frac{1-x}{1+x}\right) f\left(\frac{1+x}{1-x}\right) = f\left(\frac{1-x}{1+x}\right) + f\left(\frac{1+x}{1-x}\right)$

(where $x \neq 0, \pm 1$) and $f(3) = 28$ then the

value of $\frac{1}{605} \left(\sum_{n=1}^{10} (f(n) - 1) \right)$ is

157. If $f^2(x) f\left(\frac{1-x}{1+x}\right) = x^3 \quad [x \neq -1 \text{ and } f(x) \neq 0]$ then the value of $|[f(-2)]|$ is

(where $[.]$ is G.I.F)

158. If $f^3(x) - 3f^2(x) + 3f(x) - 1 = x^6$ then the value of $f(0)$ is

159. If $f\left(x + \frac{1}{x}\right) = x^3 + \frac{1}{x^3} - 4\left(x^2 + \frac{1}{x^2}\right) + 13$

then the value of $f(2 + \sqrt{3}) =$

SUBJECTIVE QUESTIONS

160. Find the natural number 'a' for which

$\sum_{k=1}^n f(a+k) = 16(2^n - 1)$ where the function f satisfies the relation $f(x+y) = f(x)f(y)$ for all natural numbers x, y & further $f(1) = 2$.

[IIT - 1992]

161. Let $\{x\}$ & $[x]$ denotes the fractional and integral part of a real number x respectively. Solve $4\{x\} = x + [x]$. [IIT - 1994]

162. Find the domain and range of the real

function $\frac{x+2}{x^2 - 8x - 4}$ [REE - 1995]

163. A function $f: R \rightarrow R$, where R is the set of real numbers, is defined by, $f(x)$

$= \frac{\alpha x^2 + 6x - 8}{\alpha + 6x - 8x^2}$. Find the interval of values

of α for which f is onto. Is the function one-to-one for $\alpha = 3$? Justify your answer.

[IIT - 1996]

164. Let $f(x) = Ax^2 + Bx + C$ where A, B, C are real numbers. Prove that if $f(x)$ is an integer whenever x is an integer, then the numbers $2A, A+B$ and C are all integers. Conversely, prove that if the numbers $2A, A+B$ and C are all integers then $f(x)$ is an integer whenever x is an integer.

[IIT - 1998]

KEY - LEVEL-V**SINGLE ANSWER QUESTIONS**

- | | | | |
|-------|-------|-------|-------|
| 1) a | 2) d | 3) d | 4) d |
| 5) a | 6) c | 7) c | 8) a |
| 9) b | 10) c | 11) c | 12) b |
| 13) b | 14) d | 15) a | 16) b |
| 17) d | 18) d | 19) c | 20) b |
| 21) a | 22) d | 23) c | 24) a |
| 25) a | 26) a | 27) d | 28) c |
| 29) c | 30) d | 31) d | 32) c |
| 33) c | 34) b | 35) d | 36) b |
| 37) a | 38) a | 39) d | 40) b |
| 41) c | 42) c | 43) a | 44) c |
| 45) c | 46) a | 47) a | 48) b |
| 49) d | 50) c | 51) a | 52) a |
| 53) b | 54) b | 55) a | 56) a |
| 57) a | 58) c | 59) c | 60) a |
| 61) b | 62) b | 63) a | 64) c |
| 65) c | 66) c | 67) d | 68) b |
| 69) d | 70) a | 71) b | 72) c |
| 73) c | 74) c | 75) b | 76) b |
| 77) c | 78) d | 79) b | 80) c |
| 81) a | 82) c | 83) d | 84) c |
| 85) b | 86) c | 87) c | 88) b |
| 89) c | 90) d | 91) a | 92) a |
| 93) b | | | |

MULTIANSWER QUESTIONS

- | | | | |
|----------|----------|----------|----------|
| 94) a,c | 95) c,d | 96) D | 97) A,C |
| 98) C | 99) a,d | 100) a,c | 101) D |
| 102) a,b | 103) a,b | 104) c,d | 105) b,d |
| 106) C | 107) b,d | 108) a,d | 109) B |
| 110) D | 111) D | 112) C | 113) b,d |
| 114) a,b | 115) a,b | 116) a,d | 117) D |

COMPREHENSION QUESTIONS

- | | | | |
|--------|--------|--------|--------|
| 118) c | 119) b | 120) b | 121) b |
| 122) a | 123) c | 124) b | 125) b |
| 126) a | 127) a | 128) a | 129) c |
| 130) b | 131) a | 132) b | 133) b |
| 134) d | 135) b | 136) a | 137) c |
| 138) b | 139) a | 140) c | 141) a |
| 142) b | 143) d | 144) c | 145) b |

- 146) a 147) b 148) c

MATRIX - MATCHING QUESTIONS

- 149) a-r, b-s, c-q, d-p
-
- 150) a-p,r,s; b-q; c-q; d-p,r,s

ASSERTION - REASON QUESTIONS

- 151) a 152) b 153) a 154) a
-
- 155) a

INTEGER QUESTIONS

- 156) 5 157) 2 158) 1 159) 9

SUBJECTIVE QUESTIONS

160. $a = 3$

161. $x = 0$ OR $5/3$

162. $D_f : \mathbb{R} - \left\{ (4 + 2\sqrt{5}), (4 - 2\sqrt{5}) \right\}$

$$R_f : \left(-\infty, -\frac{1}{4} \right] \cup \left[-\frac{1}{20}, \infty \right)$$

163. $\alpha = \phi$, as domain is $\mathbb{R} \Rightarrow \alpha < -\frac{9}{8}$

P a can not be 3,

Note: If $f : \mathbb{R} - \{x : 8x^2 - 6x - \alpha = 0\}$,
and $x \in \mathbb{R} \rightarrow \mathbb{R}$, then $\alpha \in [2, 14]$ for
onto function, for $\alpha = 3$, f is not one to one
164.

HINTS - LEVEL-V**SINGLE ANSWER QUESTIONS**

1. $f'(x) = 3x^2 - 6x + 6 = 3(x^2 - 2x + 2) > 0$
Disc < 0 and leading coeff > 0
f is one-one
 $f(x)$ is odd degree polynomial, $f(x)$ is onto

2.
$$f(x) = \frac{(x+1)^2 + 4}{\left(x + \frac{1}{2}\right)^2 + \frac{3}{4}} > 0$$

range $\neq$ codomain

$f(x)$ is into

clearly $f^{-1}(x) = 0$ is possible for some 'x'

$\Rightarrow f(x)$ is many-one

$$3. \quad f(0) = \frac{30}{18} = \frac{5}{3} = \frac{x^2 + 4x + 30}{x^2 - 8x + 18} \Rightarrow x = 0, x = 26$$

$f(0) = f(26) \rightarrow f$ is not one-one

$$\text{Let } y = \frac{x^2 + 4x + 30}{x^2 - 8x + 18}$$

$$\Rightarrow (y-1)x^2 - 4x(2y+1) + 18y - 30 = 0$$

Since 'x' is real, $\text{Disc} \geq 0$

$$y^2 - 32y \leq 0 \Rightarrow y \in [16 - \sqrt{243}, 16 + \sqrt{243}]$$

Not onto

$$4. \quad \text{Clearly } f(x) = 0 \text{ for all } x \leq 0 \text{ and } f(x) > 0 \text{ for all } x > 0$$

Neither one-one nor onto

$$5. \quad f^{-1}(x) = 2 + \cos x > 0 \text{ ---} \rightarrow f \text{ is one-one}$$

$$f(-\infty) = -\infty \text{ and } f(\infty) = \infty \text{ ---} \rightarrow \text{range}$$

$$= (-\infty, \infty) \text{ ---} \rightarrow f \text{ is onto}$$

$$6. \quad \text{Since composition of two continuous functions is continuous}$$

Let $h(x) = |x|$ then

$$g(x) = |f(x)| = h(f(x))$$

$\therefore g$ is continuous if f is continuous

$$7. \quad \text{Since } f(x) \text{ and } g(x) \text{ are the mirror images of each other about the line } y = a \Rightarrow f(x) \text{ and } g(x) \text{ are at equal distances from the line } y = a. \text{ Let for some particular } x_0$$

$$f(x_0) = a + k; g(x_0) = a - k$$

$h(x_0) = 2a \text{ ---} \rightarrow 'h' \text{ is a constant function ---} \rightarrow \text{many-one and into}$

$$8. \quad \text{Given condition is possible if } f(x) \text{ is constant function}$$

$\therefore f(x) = f(2) = 3$ Hence $f(x)$ is always an even function.

$$9. \quad f(-x) = \ln(-x + \sqrt{1+x^2})$$

$$= \ln \left(\left(\sqrt{1+x^2} - x \right) \frac{\sqrt{1+x^2} + x}{\sqrt{1+x^2} + x} \right)$$

$$= \ln \left(\frac{1}{x + \sqrt{1+x^2}} \right)$$

$$= -f(x)$$

$f(x)$ is an odd function

$$10. \quad a \rightarrow f(x) = x^3 - 3x^2 + 3x + 3$$

$$\Rightarrow f^{-1}(x) = 3x^2 - 6x + 3$$

$$= 3(x^2 - 2x + 1) \geq 0 \forall x \in R$$

$$b \rightarrow f(x) = 2x^3 - 3x^2 - 12x + 6$$

$$\Rightarrow f^{-1}(x) = 6x^2 - 6x - 12$$

$$= 6(x^2 - x - 2)$$

$$= 6(x^2 - 2x - 2) = 6(x-2)(x+1)$$

$$c \rightarrow \text{Let } f(x) = 3x^2 - 2x + 1$$

$$\Rightarrow f^{-1}(x) = 6x - 2$$

$$f^{-1}(x) > 0 \Rightarrow 6x - 2 > 0$$

$$\Rightarrow 3x - 1 > 0 \Rightarrow x \in \left(\frac{1}{3}, \infty \right)$$

$$f^{-1}(x) \geq 0 \Rightarrow x \in (-\infty, -1] \cup [2, \infty)$$

$$d \Rightarrow f^{-1}(x) = 3x^2 + 12x = 3(x^2 + 4x)$$

$$\Rightarrow f^{-1}(x) > 0 \Rightarrow x(x+4) > 0$$

$$\Rightarrow x \in (-\infty, -4) \cup (0, \infty)$$

$$11. \quad \text{Clearly } f(x) = 0$$

$$12. \quad f(x) > 0 \rightarrow x^2 + (k-1)x + 9 > 0 \rightarrow \text{Disc} < 0$$

$$\Rightarrow (k-1)^2 - 4(9) < 0$$

$$\Rightarrow k^2 - 2k + 1 - 36 < 0$$

$$k^2 - 2k - 35 < 0$$

$$k \in (-5, 7)$$

$$13. \quad f(-x) = \frac{\cos(-x)}{\left[\frac{-x}{\pi} \right] + \frac{1}{2}} = \frac{\cos x}{\left[\frac{x}{\pi} \right] - 1 + \frac{1}{2}} = -f(x) \text{ odd function}$$

$$14. \quad f(x) \text{ is an odd function}$$

$$\Rightarrow \left[\frac{x^2}{a} \right] = 0 \forall x \in [-10, 10]$$

$$0 \leq \frac{x^2}{a} < 1 \Rightarrow a > 100$$

$$15. \quad f^{-1}(x) = \log\left(\frac{1-x}{1+x}\right)$$

$f^{-1}(x)$ is an odd function

$f(x)$ is an even function

$$16. \quad f(x_1) = f(x_2) \Rightarrow \frac{x_1 - m}{x_1 - n} = \frac{x_2 - m}{x_2 - n}$$

$$\Rightarrow x_1 = x_2 \Rightarrow f \text{ is one-one}$$

$$\text{Let } f(x) = y \Rightarrow \frac{x-m}{x-n} = y \Rightarrow x = \frac{m-ny}{1-y},$$

Not valid for $y=1 \Rightarrow$ not onto

$$17. \quad \text{Domain of } \frac{1}{x} \text{ is } \mathbb{R} - \{0\}$$

$$\frac{1}{\sqrt{x-2}} \text{ is } x-2 > 0$$

$$\Rightarrow x > 2$$

$$\text{Domain of } 2^{\sin^{-1}x} \text{ is } [-1, 1]$$

No common region

$$18. \quad x+3 > 0 \quad \text{and} \quad (x+1)(x+2) \neq 0$$

$$\Rightarrow x \in (-3, \infty) - \{-1, -2\}$$

$$19. \quad \text{Case(i)} \quad 0 < |x| - 1 < 1 \Rightarrow 1 < |x| < 2 \Rightarrow x \in (-2, 2)$$

$$\text{and } x \in (-\infty, -1) \cup (1, \infty) \dots (1)$$

$$\Rightarrow x^2 + 4x + 4 \leq 1 \Rightarrow x^2 + 4x + 3 \leq 0$$

$$\Rightarrow -3 \leq x \leq -1 \dots (2)$$

from (1) and (2)

$$\therefore x \in (-2, -1)$$

$$\text{Case (ii)} \quad |x| - 1 > 1 \Rightarrow |x| > 2$$

$$\text{then } x^2 + 4x + 4 \geq 1 \Rightarrow x^2 + 4x + 3 \geq 0$$

$$\Rightarrow x \geq -1 \text{ or } x \leq -3$$

$$\therefore x \in (-\infty, -3] \cup (2, \infty)$$

$$\therefore x \in (-\infty, -3] \cup (-2, -1) \cup (2, \infty)$$

$$20. \quad 0 \leq 2x+3 \leq 1 \Rightarrow -3 \leq 2x \leq -2 \Rightarrow \frac{-3}{2} \leq x \leq -1$$

$$21. \quad \sin^{-1}(\log_2 x) \geq 0 \text{ and } -1 \leq \log_2 x \leq 1$$

$$\Rightarrow \log_2 x \geq 0$$

$$\Rightarrow x \geq 1$$

$$\Rightarrow -\log_2^2 \leq \log_2 x \leq \log_2^2 \Rightarrow \frac{1}{2} \leq x \leq 2$$

$$x > 0$$

$$\therefore Df = [1, 2]$$

$$22. \quad -1 \leq [x] \leq 1 \quad -1 \leq x < 2$$

$$23. \quad \frac{\log_{0.3} |x-2|}{|x|} \geq 0 \text{ and}$$

$$|x-2| \leq 1 \Rightarrow x \in [1, 2) \cup (2, 3]$$

$$24. \quad \text{Use the definition of logarithm several times}$$

$$25. \quad -1 \leq 2x \leq 1$$

$$x \in \left[-\frac{1}{2}, \frac{1}{2}\right] \Rightarrow \sin^{-1}(2x) + \frac{\pi}{6} \geq 0$$

$$\sin^{-1}(2x) \geq -\frac{\pi}{6} \Rightarrow 2x \geq -\frac{1}{2} \Rightarrow x \geq -\frac{1}{4}$$

$$\therefore \text{Domain} = \left[-\frac{1}{4}, \frac{1}{2}\right]$$

$$26. \quad -1 \leq \frac{4}{3+2\cos x} \leq 1$$

$$\Rightarrow \frac{4}{3+2\cos x} \leq 1 \Rightarrow \cos x \geq \frac{1}{2}$$

$$\Rightarrow x \in \left[2n\pi - \frac{\pi}{3}, 2n\pi + \frac{\pi}{3}\right]$$

$$27. \quad \text{There is no intersection of the values of 'x'}$$

$$28. \quad -1 \leq [2-4x^2] \leq 1$$

$$\Rightarrow -1 \leq 2-4x^2 < 2$$

$$\Rightarrow x^2 \leq \frac{3}{4}$$

$$\Rightarrow x \in \left[-\frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{2}\right]$$

but at $x=0$, $f(x)$ is not defined

$$29. \quad \text{Given that}$$

$$0 < x < 1 \Rightarrow 0 < e^x < 1 \text{ and } 0 < \ln|x| < 1$$

$$\Rightarrow x \in (-\infty, 0) \text{ and } 1 < |x| < e$$

$$x \in (-3, -1) \cup (1, e)$$

Domain is $(-e, -1)$

- 30) $\cos(\sin x) \geq 0$ is true for all $x \in R$; $\log \{x\}_x \geq 0$
 case(i) $0 < x < 1 \Rightarrow \{x\} \leq 1 \Rightarrow x \in (0, 1)$, $x > 1$
 and $\{x\} \leq 1$ not possible

- 31) $\because \frac{1}{\sqrt{-x^2}}$ is not defined for any x Domain $= \phi$

32. $-1 \leq [2 - 3x^2] \leq 1$

$$-1 \leq 2 - 3x^2 < 2$$

$$0 < x^2 \leq 1$$

$$x \in [-1, 1] - \{0\}$$

33. $-1 \leq \frac{1+x^2}{2x} \leq 1$

$$\left| \frac{1+x^2}{2x} \right| \leq 1$$

$$1+x^2 \leq 2|x|$$

$$|x|^2 - 2|x| + 1 \leq 0$$

$$(|x| - 1)^2 \leq 0$$

$$|x| = 1$$

$$x = \pm 1$$

$$D_f = \{-1, 1\}$$

34. $-1 \leq \sec(\cos^{-1} x) \leq 1$ and $-1 \leq \operatorname{cosec}(\sin^{-1} x) \leq 1$

$$\sec(\cos^{-1} x) = \pm 1 \text{ and } \operatorname{cosec}(\sin^{-1} x) = \pm 1$$

$$\cos^{-1} x = 0 \text{ or } \pi \text{ and } \sin^{-1} x = \frac{\pi}{2}, \frac{-\pi}{2}$$

$$\text{Domain} = \{-1, 1\}$$

- 35) $\because$ Range of $\log x$ is R i.e $\log x$ will take the values

$$\text{from } -\frac{\pi}{2} \text{ to } \frac{\pi}{2} \text{ also}$$

$$\Rightarrow \text{range}(-\infty, \infty)$$

- 36) Let

$$y = \frac{e^{-x}}{1+[x]} \Rightarrow (1+[x])y = e^{-x} > 0 \Rightarrow (1+[x])y > 0$$

$$\Rightarrow 1+[x] < 0; y < 0 \text{ (or)} 1+[x] > 0; y > 0$$

$$\therefore R_f = R - \{0\}$$

- 37) Use $A.M \geq G.M$

38) $f(x) = \left(\sin x - \frac{5}{2} \right)^2 - \frac{49}{4}$ where

$$\frac{9}{4} \leq \left(\sin x - \frac{5}{2} \right)^2 \leq \frac{49}{4}$$

39) $f(x) = \frac{\{x\}}{1+\{x\}} \Rightarrow \{x\} = \frac{y}{1-y}$

$$\Rightarrow 0 \leq \frac{y}{1-y} < 1 \Rightarrow y \in \left[0, \frac{1}{2} \right)$$

- 40) Here $S = \text{co-domain} = \text{onto}$

$\because f$ is onto
use

$$a \sin x + b \cos x + c \in \left[c - \sqrt{a^2 + b^2}, c + \sqrt{a^2 + b^2} \right]$$

- 41) $\lambda = \frac{1}{4}$ is the only solution

- 42) $7-x > 0$ and $x-3 \geq 0$ and
 $7-x \geq x-3 \Rightarrow x \in \{3, 4, 5\}$ is the domain

$$\text{Now } x=3 \Rightarrow 4_{P_0} = 1$$

$$x=4 \Rightarrow 3_{P_1} = 3, x=5 \Rightarrow 2_{P_2} = 2$$

- 43) Let

$$y = \frac{x^2 - x + 1}{x^2 + x + 1} \Rightarrow yx^2 + yx + y - x^2 + x - 1 = 0$$

$$\Rightarrow (y-1)x^2 + (y+1)x + (y-1) = 0$$

$$\text{Now } \Delta \geq 0 \Rightarrow y \in \left[\frac{1}{3}, 3 \right]$$

- 44) $0 \leq \{x\} < 1$

$$\Rightarrow \{x\} = 0 \text{ is ruled out Now } 0 < \{x\} < 1$$

$$\Rightarrow \sin 0 < \sin \{x\} < \sin 1$$

$$\Rightarrow \frac{1}{\sin \{x\}} > 1.18, \therefore \left[\frac{1}{\sin \{x\}} \right] \geq 1$$

- 45) Since Nr is of the form $\tan(n\pi)$ which is zero

- 46) Let

$$y = f(x) = \sqrt{a-x} + \sqrt{x-b} \Rightarrow \text{clearly } y \geq 0$$

$$y^2 = (a-x) + (x-b) + 2\sqrt{a-x}\sqrt{x-b}$$

$$y^2 = (a-b) + 2\sqrt{\left(\frac{a-b}{2}\right)^2 - \left(x - \left(\frac{a+b}{2}\right)\right)^2}$$

the greatest value of

$$y^2 = (a-b) + 2\left(\frac{a-b}{2}\right) = 2(a-b)$$

the least value of

$$y^2 = a-b \Rightarrow \text{range of}$$

$$f(x) = \left[\sqrt{a-b}, \sqrt{2(a-b)} \right]$$

47) as $x^3 + 3x^2 + 10x$ is a cubic polynomial

$$\therefore -\infty < x^3 + 3x^2 + 10x < \infty$$

$$\Rightarrow -\infty < x^3 + 3x^2 + 10x + 2\sin x$$

$$< \infty \because -1 \leq \sin x \leq 1$$

48. min.value of $f(x) = 2c^2 - b^2$

$$\text{max.value of } g(x) = b^2 + c^2$$

$$\text{Now } 2c^2 - b^2 > b^2 + c^2$$

$$\Rightarrow c^2 > 2b^2$$

$$\Rightarrow |c| > |b|\sqrt{2}$$

49. Let $y = f(x) = \cot^{-1}(2-x^2)$

$$y \in (0, \pi) \text{-----(1)}$$

$$\cot y = 2x - x^2$$

$$(x-1)^2 = 1 - \cot y \geq 0$$

$$\cot y \leq 1$$

$$y \geq \pi/4 \text{-----(2)}$$

$$\text{from (1) and (2) } R_f = \left[\frac{\pi}{4}, \pi \right)$$

50. Clearly D_f is $\left\{ -\frac{1}{2}, \frac{1}{2} \right\}$

$$f(x) \text{ is minimum when } x = \frac{1}{2}$$

$$\text{and it is } f\left(\frac{1}{2}\right) = \frac{\pi}{2}$$

$f(x)$ is maximum when $x = \frac{1}{2}$ and it is

$$\left(\frac{-1}{2}\right) = \frac{3\pi}{2}$$

$$\therefore \frac{\pi}{2} + \frac{3\pi}{2} = 2\pi$$

51. As we know

$$\left(\frac{a+b+c+d}{4}\right)^2 \leq \frac{a^2+b^2+c^2+d^2}{4}$$

(using tchebycheff's inequality)

$$\left(\frac{8-e}{4}\right)^2 \leq \frac{16-e^2}{4} \rightarrow e(5e-16) \leq 0$$

$$\rightarrow e \in \left[0, \frac{16}{5}\right]$$

52) $-1 \leq \cos^{-1}[x] \leq 1$ and $-1 \leq [x] \leq 1$

$$\Rightarrow x \in [1, 2) \text{ and clearly range is } \{0\}$$

53) $\frac{|x|}{x} > 0$ and $[x-1] > 0$ and $[x-1] \neq 1$

$$[x] > 1 \quad [x]-1 \neq 1$$

$$[x] \neq 2 \Rightarrow x \in [3, \infty)$$

$$\text{clearly } \frac{|x|}{x} = 1 \text{ only } \Rightarrow \text{range} = \log 1 = 0$$

54) Range of $f(x)$ for which $g(f(x))$ is defined is $\{3, 4\}$. Hence domain of $g(f(x))$ has two elements $\Rightarrow$ range of $g(f(x))$ has two elements.

55) $f^1(x) = 3x^2 + 6x + 4 + a \cos x - b \sin x \geq 0$

leading coeff > 0

$$\Rightarrow 3x^2 + 6x + 4 \geq b \sin x - a \cos x$$

$$\Rightarrow 3x^2 + 6x + 4 \geq \sqrt{a^2 + b^2}$$

$$\Rightarrow 3(x+1)^2 + 1 \geq \sqrt{a^2 + b^2}$$

max.value of $a^2 + b^2$ is 1

56) $\frac{\sqrt{4-x^2}}{1-x} > 0$; $4-x^2 > 0 \Rightarrow x \in (-2, 1)$ and

range is $[-1, 1]$

57) Let f be not one-one

$$\text{i.e } x_1 \neq x_2 \Rightarrow f(x_1) = f(x_2)$$

$$\Rightarrow g(f(x_1)) = g(f(x_2)) \Rightarrow \text{gof is not one-}$$

one, which is a contradiction

$$58. \quad g(x) = 1 + \{x\} \Rightarrow g(x) \geq 1 \text{ Now } f(f(x)) = 1$$

$$59. \quad \begin{aligned} f(f(x)) &= 1 + f(x); 0 \leq f(x) \leq 2; 0 \leq x \leq 2 \\ &= 3 - f(x); 2 < f(x) \leq 3; 2 < x \leq 3 \end{aligned}$$

$$\begin{aligned} \text{Now } (f \circ f)(x) &= 2 + x; -1 \leq x \leq 1; 0 \leq x \leq 2 \\ &= 2 - x; 1 < x \leq 2; 0 \leq x \leq 2 \\ &= 4 - x; 1 \leq x \leq 3; 2 < x \leq 3 \\ &= -x; 0 \leq x < 1; 2 < x \leq 3 \end{aligned}$$

$$\begin{aligned} \text{Now } (f \circ f)(x) &= 2 + x; 0 \leq x \leq 1 \\ &= 2 - x; 1 < x \leq 2 \\ &= 4 - x; 2 < x \leq 3 \end{aligned}$$

$$60. \quad f(f(x)) = x; \text{ also } g(x) = x^2 + px + q$$

$$\therefore g(x) - x = 0 \Rightarrow x^2 + (p-1)x + q = 0$$

As it has imaginary roots,

$$\Delta < 0 \Rightarrow (p-1)^2 - 4q < 0$$

and coeff x^2 is already +ve

$$\Rightarrow x^2 + (p-1)x + q > 0 \quad \forall x \in R$$

$$(i.e) \quad g(x) - x > 0 \Rightarrow g(g(x)) - f(f(x))$$

$$\Rightarrow g(g(x)) - g(x) > 0$$

$$\Rightarrow g(g(x)) - x$$

$$= [g(g(x)) - g(x)] + [g(x) - x] > 0$$

has no real roots

$$61. \quad f(g(x)) = f(1 + \sqrt{x})$$

$$\therefore f(1 + \sqrt{x}) = 3 + 2\sqrt{x} + x$$

$$\text{Let } 1 + \sqrt{x} = y$$

$$x = (y-1)^2$$

$$\therefore f(y) = 3 + 2(y-1) + (y-1)^2$$

$$= 2 + y^2$$

$$= f(x) = 2 + x^2$$

$$62. \quad g(f(x)) = (\sin x + \cos x)^2 - 1 = \sin(2x)$$

Clearly $g(f(x))$ is invertible in

$$\left[\frac{-\pi}{2}, \frac{\pi}{2} \right] \rightarrow x \in \left[\frac{-\pi}{4}, \frac{\pi}{4} \right]$$

63. as f^{-1} is the image of 'f' under the line mirror $y=x$, f^{-1} and f have same monotonic nature
 $\Rightarrow f^{-1}$ is also decreasing function

Now let

$$z = f^{-1}(-x)$$

$$-x = f(z)$$

$$= f(-z) \quad (\because f \text{ is odd function})$$

$$-z = f^{-1}(x)$$

$$z = -f^{-1}(x)$$

$$f^{-1}(-x) = -f^{-1}(x)$$

f^{-1} is an odd function

64. Clearly $f^{-1}(x) = \sqrt{x+1} - 1$ Now

$$f(x) = f^{-1}(x) \rightarrow (x+1)^2$$

$$= \sqrt{x+1} - 1 \rightarrow x = 0 \text{ or } -1$$

65. Let T be the period of f(x), then

$$f(x+T) = \sin(x+T) + \cos(ax+T)$$

$$= f(x) = \sin x + \cos(ax)$$

$$\sin(x+T) + \cos(ax+T)$$

$$= \sin x + \cos(ax)$$

put $x=0$ and $x=-T$ respectively

$$\sin T + \cos(aT) = 1$$

$$-\sin T + \cos(aT) = 1 \quad \text{solve these equations}$$

$$\sin(T) = 0 \text{ and } \cos(aT) = 1$$

$$T = n\pi; n \in \mathbb{Z} \quad aT = 2m\pi; m \in \mathbb{Z}$$

$$\frac{aT}{T} = \frac{2m\pi}{n\pi} = \frac{2m}{n}, \text{ a rational number}$$

66. Period of $|\sin(4x)| + |\cos(4x)|$ is $\frac{\pi}{8}$

Period of

$$|\sin(4x) - \cos(4x)| + |\sin(4x) + \cos(4x)|$$

$$\text{is } \frac{\pi}{8}$$

$$\therefore \text{period} = \pi/8$$

$$67. f(x) = \cos x + (x - [x])$$

Here period of $\cos x$ is 2π and the period of $x - [x]$ is 1

But L.C.M of 2π and 1 does not exist

$$68. f(x) = \sum_{r=1}^n \left[\sin\left(\frac{x}{2^{r-1}}\right) + \tan\left(\frac{x}{2^r}\right) \right]$$

periods are $2\pi, 2\pi, 8\pi, 8\pi, 32\pi, 32\pi, 32\pi, \dots, 2^n\pi, 2^n\pi$

$$\text{i.e. } L.C.M = 2^n\pi$$

$$69. f(x+2\pi) = \sin(\cos(2\pi+x) + (2\pi+x)) \\ = \sin(2\pi+x+\cos x) = \sin(x+\cos x) = f(x) \\ \Rightarrow \text{period of } f(x) \text{ is } 2\pi$$

$$70. \text{period of } x - [x] \text{ is } 1$$

$$\text{period of } |\cos(\pi x)| \text{ is } \frac{\pi}{\pi} = 1$$

$$\text{period of } |\cos(2\pi x)| \text{ is } \frac{\pi}{2\pi} = \frac{1}{2} \dots\dots$$

$$L.C.M \text{ of } 1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n} \text{ is } 1$$

$$71. \text{The period of } \left| \sin^3\left(\frac{x}{2}\right) \right| \text{ is } 2\pi$$

$$\text{The period of } \left| \cos^5\left(\frac{x}{5}\right) \right| \text{ is } 5\pi$$

$$L.C.M \text{ of } 2\pi, 10\pi \text{ is } 10\pi$$

$$72. \text{From given data, } g(x) \text{ must be linear function} \\ g(x) = ax + b$$

$$\text{Also } g(2) = 3 \Rightarrow 2a + b = 3; g(4) = 7$$

$$\Rightarrow 4a + b = 7 \text{ solving } a = 2, b = -1$$

$$\therefore g(x) = 2x - 1 \Rightarrow g(6) = 11$$

$$73. f(x+p)$$

$$= 1 + \left[2 - 3f(x) + 3(f(x))^2 - (f(x))^3 \right]^{1/3}$$

$$f(x+p) = 1 + \left[1 + (1 - f(x))^3 \right]^{1/3}$$

$$f(x+p) - 1 = \left[1 - (f(x) - 1)^3 \right]^{1/3}$$

$$\text{Let } f(x) - 1 = g(x)$$

$$g(x+p) = \left[1 - g(x)^3 \right]^{1/3} \dots\dots(1)$$

replace 'x' by $x+p$

$$g(x+2p) = \left[1 - (g(x+p))^3 \right]$$

$$g(x+2p) = \left[1 - (1 - g(x))^3 \right]^{1/3} \text{ (from (1))}$$

$$g(x+2p) = g(x) \forall x \in R$$

$$f(x+2p) - 1 = f(x) - 1$$

$$f(x+2p) = f(x)$$

period of 'f' is $2p$

74. Conceptual

75. Since we know that if $f(x)$ is a periodic function, then

$f(ax+b)$ is also periodic function $\rightarrow g(x)$ is a linear function of 'x'

$$f(x+2b) = f(-x) \text{ again replace } x \text{ by } -x$$

$$\text{then } f(x) = f(2b-x) = f(2b+x)$$

$$[\because f(a-x) = f(a+x) \text{ and } a > b. \text{ Here } a = 2b]$$

$$f(2b+x) = f(x) \rightarrow \text{period of } f(x) \text{ is } 2b$$

76. Conceptual

$$77) \text{ Given } 2f(\sin x) + f(\cos x) = x \dots\dots(i)$$

$$\text{Replacing } x \text{ by } \left(\frac{\pi}{2} - x\right) \text{ in equation (i),}$$

we get

$$2f\left(\sin\left(\frac{\pi}{2} - x\right)\right) + f\left(\cos\left(\frac{\pi}{2} - x\right)\right) = \frac{\pi}{2} - x$$

$$\text{or } 2f(\cos x) + f(\sin x) = \frac{\pi}{2} - x \dots\dots(ii)$$

Multiplying in equation (i) by 2, then

$$4f(\sin x) + 2f(\cos x) = 2x \quad \dots(iii)$$

Now subtraction Eq. (ii) from Eq. (iii), we get

$$3f(\sin x) = 2x - \left(\frac{\pi}{2} - x\right) = 3x - \frac{\pi}{2}$$

$$\text{or } f(\sin x) = x - \frac{\pi}{6}$$

$$\text{or } f(x) = \sin^{-1} x - \frac{\pi}{6}$$

$$\text{Hence, } f(x) = \sin^{-1} x - \frac{\pi}{6} \quad \forall x \in [-1, 1].$$

78. By observation, it is easy to decide that

$$f(x) = \ln x$$

$$79. f(x - f(y)) = f(f(y)) + x.f(y) + f(x) - 1 \quad \dots(1)$$

$$\text{put } x = f(y) = 0$$

$$\rightarrow f(0) = f(0) + 0 + f(0) - 1$$

$$\Rightarrow f(0) = 1 \dots(2)$$

$$\text{Again put } x = f(y) = \lambda \text{ in (1)}$$

$$f(0) = f(\lambda) + \lambda^2 + f(\lambda) - 1 \dots(1)$$

$$\Rightarrow 1 = 2f(\lambda) + \lambda^2 - 1$$

$$f(\lambda) = \frac{2 - \lambda^2}{2}$$

$$= 1 - \frac{\lambda^2}{2}$$

$$\Rightarrow f(x) = 1 - \frac{x^2}{2}$$

$$80. f(x + y) - kxy = f(x) + 2y^2; \text{ replace 'x' by '-x', then we get}$$

$$f(0) + kx^2 = f(x) + 2x^2$$

$$\Rightarrow f(x) = f(0) + kx^2 - 2x^2$$

$$\text{put } x = 1$$

$$f(1) = f(0) + k - 2 = 2$$

$$\Rightarrow f(2) = f(0) + 4k - 8 = 8$$

$$\text{solving } k = 4 \text{ and } f(0) = 0$$

$$\therefore f(x) = 2x^2$$

$$81. f(x + y) = f(x).f(y) \dots\dots(1)$$

$$\text{put } x = y = 0 \Rightarrow f(0) = 1 [\because f(0) \neq 0]$$

$$\text{put } y = -x \text{ in (1) then}$$

$$f(0) = f(x).f(y) \rightarrow 1 = f(x).f(-x)$$

$$\Rightarrow f(-x) = \frac{1}{f(x)} \dots\dots(2)$$

$$\text{Now } F(x) = \frac{f(x)}{1 + (f(x))^2}$$

$$\Rightarrow F(-x) = \frac{f(-x)}{1 + (f(-x))^2}$$

$$= \frac{\frac{1}{f(x)}}{1 + \frac{1}{(f(x))^2}} \text{ from (2)}$$

$$= F(x) \Rightarrow F \text{ is an even function}$$

$$82. \text{put } y = \frac{1}{x} \rightarrow 2 + f(x).f\left(\frac{1}{x}\right)$$

$$= f(x) + f\left(\frac{1}{x}\right) + f(1) \dots\dots(1)$$

$$\text{put } x = 1$$

$$2 + (f(1))^2 = 3f(1) \rightarrow f(1) = 1 \text{ (or) } 2$$

$$\text{but } f(1) \neq 1, \text{ otherwise, from the given relation}$$

$$2 + f(x)f(1) = f(x) + f(1) + f(x) \quad (\text{or})$$

$$f(x) = 1, \text{ which is not possible as } f(2) = 5$$

$$\text{Hence } f(1) = 2 \text{ from (1)}$$

$$f(x).f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$$

$$\rightarrow f(x) = \pm x^n + 1$$

$$\Rightarrow f(f(2)) = 26$$

$$83. g(x) = f(x) - 1 \dots\dots(1)$$

like this

93. Since

$$|f(x) - f(y)| \leq |x - y|^3 \text{ is true } \forall x, y \in R$$

$$\text{we have for } x \neq y, \frac{|f(x) - f(y)|}{|x - y|} \leq |x - y|^2$$

$$\lim_{y \rightarrow x} \left| \frac{f(x) - f(y)}{x - y} \right| \leq \lim_{y \rightarrow x} |x - y|^2$$

$$\rightarrow \lim_{y \rightarrow x} \left| \frac{f(x) - f(y)}{x - y} \right| \leq 0$$

$$|f'(x)| \leq 0 \Rightarrow f'(x) = 0$$

$f(x)$ is a constant function

MULTIANSWER QUESTIONS

94. $f(x) = \frac{x}{x+1} = 1 - \frac{1}{1+x}$

$$\Rightarrow f'(x) = \frac{1}{(1+x)^2} > 0 \Rightarrow 'f'$$

is increasing function

f is one-one

Now $\text{range} = [f(0), f(\infty)) = [0, 1) \neq R$ into function

95. For all $x \in (e, \infty) \Rightarrow \log x > 1$

$$\log(\log x) > 0 \text{ for all } x \in (e, \infty)$$

$$f'(x) = \frac{1}{\log(\log x)} \cdot \frac{1}{\log x} \cdot \frac{1}{x} > 0$$

$$\log(\log(\log x)) \in (-\infty, \infty) \text{ for all } x \in (e, \infty)$$

$\Rightarrow f$ is increasing function

96. We are given that

$$f: R \rightarrow R \text{ s.t. } f(x) = \begin{cases} 0, & x \in \text{rational} \\ x, & x \in \text{irrational} \end{cases}$$

$$g: R \rightarrow R \text{ s.t. } g(x) = \begin{cases} 0, & x \in \text{irrational} \\ x, & x \in \text{rational} \end{cases}$$

$$(f - g)(x) = \begin{cases} -x & \text{if } x \in \text{rational} \\ x & \text{if } x \in \text{irrational} \end{cases}$$

FUNCTIONS

Since $f, g: R \rightarrow R$ for any x there is only one value of $(f(x) - g(x))$ whether x is rational or irrational. Moreover as $x \rightarrow R$, $f(x) - g(x)$ also belong to R . Therefore $(f - g)$ is one-one onto.

(D) is the correct option.

97. (A, C)

$$f(x) = \cos[\pi^2]x + \cos[-\pi^2]x$$

We know $9 < \pi^2 < 10$ and $-10 < -\pi^2 < -9$

$$[\pi^2] = 9 \text{ and } [-\pi^2] = -10$$

$$\Rightarrow f(x) = \cos 9x + \cos(-10x)$$

$$f(x) = \cos 9x + \cos(-10x) \Rightarrow f(x) = \cos 9x + \cos 10x$$

$$(A) f\left(\frac{\pi}{2}\right) = \cos \frac{9\pi}{2} + \cos 5\pi = -1 \text{ (True)}$$

$$(B) f(\pi) = \cos 9\pi + \cos 10\pi = -1 + 1 = 0 \text{ (False)}$$

$$(C) f(-\pi) = \cos(-9\pi) + \cos(-10\pi) = \cos 9\pi + \cos 10\pi = -1 + 1 = 0 \text{ (True)}$$

$$(D) f\left(\frac{\pi}{4}\right) = \cos \frac{9\pi}{4} + \cos \frac{5\pi}{2} = \cos$$

$$\left(2\pi + \frac{\pi}{4}\right) + 0 \text{ (False)}$$

Thus (A) and (C) are the correct options.

98. Let $h(x) = |x|$ then $g(x) = |f(x)| = h(f(x))$

Since composition of two continuous function is continuous, g is continuous if f is continuous.

99. Let $f(x) = [x]^2 + [x] + 1 - 3$

$$= [x]^2 + [x] - 2 = ([x] + 2)([x] - 1)$$

$$\text{Clearly } f(x) = 0$$

$$[x] = -2 \text{ (or)} [x] = 1$$

f is many-one and range contains only

integers $\Rightarrow \text{range} \neq \text{co-domain} \Rightarrow f$ is into

100. $\log_{\frac{1}{2}}(x^2 - 7x + 13) > 0$ and $x^2 - 7x + 13 > 0$

$$x^2 - 7x + 13 < 1$$

$$x^2 - 7x + 12 < 0$$

$$x \in (3, 4)$$

$$x^2 - 7x + 13 > 0$$

always true i.e. $\forall x \in R$

101. For domain of $f(x) = \frac{\log_2(x+3)}{x^2 + 3x + 2}$

$$x^2 + 3x + 2 \neq 0 \text{ and } x + 3 > 0$$

$$\Rightarrow x \neq -1, -2 \text{ and } x > -3$$

$$\therefore D_f = (-3, \infty) - \{-1, -2\}$$

$$102. 100x > 0 \quad \text{and} \quad 100x \neq 1 \quad \text{and}$$

$$\frac{2 \log_{10} x}{-x} > 0 \Rightarrow \frac{2 \log_{10} x + 1}{x} < 0$$

$$2 \log_{10} x + 1 < 0$$

$$\log_{10} x < \frac{-1}{2}, \quad x < 10^{\frac{-1}{2}}$$

$$103. 0 \leq 2 \sin x \leq 1 \Rightarrow 0 \leq \sin x \leq \frac{1}{2}$$

$\Rightarrow a, b, c$ are correct

$$104. 3x^2 - 4x + 5 > 0 \text{ is true } \forall x \in \mathbb{R} \because \text{Disc} < 0 \text{ and leading coeff} > 0$$

$$\text{Let } f(x) = y \Rightarrow \log_{10}(3x^2 - 4x + 5) = y$$

$$\Rightarrow 3x^2 - 4x + (5 - 10^y) = 0$$

As x is real,

$$\Delta \geq 0 \Rightarrow 10^y \geq \frac{11}{3} \Rightarrow y \in [\log_{10}\left(\frac{11}{3}\right), \infty)$$

$$105. \frac{|x|}{x} > 0; [x] > 0 \text{ and } [x] \neq 1$$

$$\Rightarrow x > 0 \quad x \in [1, \infty) \Rightarrow x \in [2, \infty) = \text{domain}$$

$$\text{For } x \in [2, \infty) \Rightarrow \log_{[x]}\left(\frac{|x|}{x}\right) = \log 1 = 0$$

$$\Rightarrow \cos^{-1}(0) = \frac{\pi}{2}$$

$$\therefore \text{range} = \left\{\frac{\pi}{2}\right\}$$

$$106. \text{ We have } f(x) =$$

$$\frac{x^2 + x + 2}{x^2 + x + 1} = \frac{(x^2 + x + 1) + 1}{x^2 + x + 1} = 1 + \frac{1}{\left(x + \frac{1}{2}\right)^2 + \frac{3}{4}}$$

We can see here that as $x \rightarrow \infty$, $f(x) \rightarrow 1$ which is the min value of $f(x)$. Also $f(x)$ is max when

$\left(x + \frac{1}{2}\right)^2 + \frac{3}{4}$ is min which is so when $x = -1/2$ and then $3/4$.

$$f_{\max} = 1 + \frac{1}{3/4} = \frac{7}{3}, \quad R_f = (1, 7/3]$$

$$107. f(x) = \sin^{-1}\left(\frac{2x}{1+x^2}\right) = 2 \tan^{-1} x; -1 \leq x \leq 1$$

$$= \pi - 2 \tan^{-1} x; x > 1$$

$$= -\pi + 2 \tan^{-1} x; x < -1$$

$$g(x) = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) = 2 \tan^{-1} x; x \geq 0$$

$$= -2 \tan^{-1} x; x < 0$$

but given that $f(x)$ and $g(x)$ are identical functions
 $\Rightarrow \text{Domain of } f(x) = \text{Domain of } g(x)$

$$108. f(x) = \sqrt{2} \sin\left(x + \frac{\pi}{4}\right) + 2\sqrt{2} \quad (\text{or})$$

$$\sqrt{2} \cos\left(x - \frac{\pi}{4}\right) + 2\sqrt{2} \Rightarrow -\frac{\pi}{2} \leq x + \frac{\pi}{4} \leq \frac{\pi}{2}$$

$$-\frac{\pi}{2} \leq x + \frac{\pi}{4} \Rightarrow -\frac{\pi}{2} - \frac{\pi}{4}$$

$$X = \left[-\frac{3\pi}{4}, \frac{\pi}{4}\right] (\text{or}) \Rightarrow X = \left[-\frac{\pi}{4}, \frac{5\pi}{4}\right]$$

$$\text{Clearly } Y = [\sqrt{2}, 3\sqrt{2}]$$

$$109. f(x) = 3x - 5 \text{ (given)}$$

$$\text{Let } y = f(x) = 3x - 5$$

$$\Rightarrow y + 5 = 3x \Rightarrow x = \frac{y+5}{3} \quad \dots (1)$$

$$\text{and } y = f(x) \Rightarrow x = f^{-1}(y) \quad \dots (2)$$

From (1) and (2)

$$f^{-1}(y) = \frac{y+5}{3} \Rightarrow f^{-1}(x) = \frac{x+5}{3}.$$

$$110. f(x) = \frac{\alpha x}{x+1}, x \neq -1$$

$$f(f(x)) = x \Rightarrow \frac{\alpha \left(\frac{\alpha x}{x+1}\right)}{\frac{\alpha x}{x+1} + 1} = x$$

$$\Rightarrow \frac{\alpha^2 x}{(\alpha+1)x+1} = x \Rightarrow (\alpha+1)x^2 + (1-\alpha^2)x = 0 \quad \dots (1)$$

$$\Rightarrow \alpha+1=0 \text{ and } 1-\alpha^2=0$$

$$\left[\begin{array}{l} \text{As true } \forall x \neq -1 \\ \therefore \text{Eq.(1) is an identity} \end{array} \right]$$

$$\Rightarrow \alpha = -1.$$

$$111. f(x) = \sin x + \cos x \quad g(x) = x^2 - 1$$

$$\Rightarrow g(f(x)) = (\sin x + \cos x)^2 - 1 = \sin 2x$$

Clearly $g(f(x))$ is invertible in $-\frac{\pi}{2} \leq 2x \leq \frac{\pi}{2}$

$$\left[\because \sin \theta \text{ is invertible when } -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2} \right]$$

$$\Rightarrow -\frac{\pi}{4} \leq x \leq \frac{\pi}{4}.$$

112. Since only (c) satisfy given definition

$$\text{i.e., } f(f^{-1}(B)) = B$$

$$\text{Only If } B \subseteq f(X).$$

113. Conceptual

$$\begin{aligned} 114. \quad f(n+2) - f(n+1) &= \underline{n+2} = (n+2)\underline{n+1} \\ &= (n+2)[f(n+1) - f(n)] \\ \Rightarrow f(n+2) &= (n+3)f(n+1) - (n+2)f(n) \\ \therefore p(x) &= x+3; Q(x) = -x-2 \end{aligned}$$

$$115. \quad (x+1) - [x+1] + 2x = 4[x+1] - 6$$

$$3x = 5[x] - 2 \dots \dots \dots (1)$$

$$3([x] + \{x\}) = 5[x] - 2$$

$$\Rightarrow 3\{x\} = 2[x] - 2 \dots \dots \dots (2)$$

but

$$0 \leq \{x\} < 1 \Rightarrow 0 \leq \{3x\} < 3 \Rightarrow 0 \leq 2[x] - 2 < 3$$

$$\Rightarrow [x] = 1, 2 \text{ Now } [x] = 1 \Rightarrow x = 1$$

$$[x] = 2 \Rightarrow x = 8/3 \text{ [from (1) and (2)]}$$

$$116. \quad f(-x) = 0 \text{ for } -x = 0$$

$$= x^2 \sin\left(\frac{-\pi}{x}\right) \text{ for } -1 \leq -x < 1 (-x \neq 0)$$

$$= -x|x| \text{ for } x \leq -1 \text{ or } x \geq 1$$

$$= -f(x) \text{ is an odd function}$$

$$f^1(x) \text{ is an even function}$$

$$117. \text{ It is given that } 2^x + 2^y = 2 \forall x, y \in \mathbb{R}$$

$$\text{Therefore, } 2^x = 2 - 2^y < 2 \Rightarrow 0 < 2^x < 2$$

$$\text{Taking log for both side with base 2}$$

$$\Rightarrow \log_2 0 < \log_2 2^x < \log_2 2$$

$$\text{Hence domain is } -\infty < x < 1.$$

COMPREHENSION QUESTIONS

$$118. \quad f(x) = |\cos x| \text{ which will be in } \left[\frac{-\pi}{2}, 0 \right]$$

$$119. \quad f(x) = |x| \Rightarrow B = [0, \infty)$$

$$120. \quad [x] + [-x] = 1; x \notin \mathbb{Z} \\ = 0; x \in \mathbb{Z} \Rightarrow B = \{-1, 0\}$$

$$121. \quad (f \circ f)^{-1}(x) = x \Rightarrow f(f^{-1}(x)) = x \rightarrow 2^{(f^{-1}(x))^2} \\ (f^{-1}(x))^4 - 4(f^{-1}(x))^2 - \log_2 x = 0$$

$$(f^{-1}(x))^2 = 2 + \sqrt{4 + \log_2 x}$$

$$\text{Range of } f^{-1}(x) = \sqrt{2 + \sqrt{4 + \log_2 x}}$$

122. As 'g' is invertible

$$g^1(x) = \frac{-\cos x}{(\sin x - 2)^2} \geq 0$$

$$\therefore x \in \left[\frac{\pi}{2}, \pi \right]$$

g is increasing function

$$\Rightarrow \text{range of } g \text{ is } \left[f\left(\frac{\pi}{2}\right), g(\pi) \right]$$

$$= [-5, -2]$$

123. Use the above information

$$124. \quad f(x) = \frac{1}{2} \left[f(xy) + f\left(\frac{x}{y}\right) \right] \dots \dots (1)$$

interchanging 'x' and 'y'

$$f(y) = \frac{1}{2} \left[f(xy) + f\left(\frac{y}{x}\right) \right] \dots \dots (2)$$

$$(1) - (2) \Rightarrow f(x) - f(y) = \frac{1}{2} \left[f\left(\frac{x}{y}\right) - f\left(\frac{y}{x}\right) \right] \\ \dots \dots (3)$$

in (1) put $x = 1$

$$f(1) = \frac{1}{2} \left[f(y) + f\left(\frac{1}{y}\right) \right] \dots \dots (4)$$

$$f(1) = 0 \Rightarrow f(y) = -f\left(\frac{1}{y}\right)$$

$$\therefore f\left(\frac{x}{y}\right) = -f\left(\frac{y}{x}\right)$$

$$\Rightarrow f(x) - f(y) = f\left(\frac{x}{y}\right)$$

$$125 \quad f^1 x = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$\Rightarrow f^1(3) = \frac{2}{3}$$

$$126 \quad f^1(x) = \frac{2}{x}$$

$$\Rightarrow f(x) = 2 \log x + c$$

$$f(1) = 0 \Rightarrow c = 0 \Rightarrow f(x) = 2 \log x$$

$$\Rightarrow f(e) = 2$$

127, 128, 129.

$$f(2-x) = f(2+x) \text{ --- (1)}$$

replace 'x' by '2-x'

$$\rightarrow f(x) = f(4-x) \text{ --- (2) Also given}$$

$$f(2-x) = f(x) \text{ --- (3)}$$

$$\text{from (2) and (3) } f(4-x) = f(20-x)$$

$$\text{Now replace 'x' by '4-x' } \rightarrow f(x) = f(x+16)$$

Hence period of $f(x)$ is 16

$$\text{If 1 is a period then } f(x) = f(x+1) \forall x \in R$$

$$\Rightarrow f(2) = f(3) = f(4) = f(5) = f(6)$$

which contradicts the given hypothesis is

$$f(2) \neq f(6)$$

$\therefore 1$ can not be the period

$$130. \quad y = f(x) = \frac{x^2}{4} - ax + a^2 + a - 2$$

$f(0)$ has opp. sign as that of leading coeff

$$f(0) < 0 \Rightarrow a^2 + a - 2 < 0$$

$$\Rightarrow -2 < a < 1 \Rightarrow a = -1, 0$$

i.e 2 values

$$131. \quad \text{vertex of } C_a \text{ is } (2a, a-2) \text{ let vertex } = (h, k)$$

$$\therefore h = 2a; k = a - 2 \Rightarrow h = 2(k + 2)$$

$$x = 2y + 4 \rightarrow x - 2y - 4 = 0$$

$$132. \quad a = 3 \Rightarrow \text{curves are } y = \frac{x^2}{4} - 3x + 10 \text{ (1)}$$

$$\& y = 2 - \frac{x^2}{4} \text{ (2)}$$

$$y = mx + c \text{ with (1)} \Rightarrow mx + c = \frac{x^2}{4} - 3x + 10$$

$$\frac{x^2}{4} - (m+3)x + (10-c) = 0$$

$$\text{Disc} = 0 \Rightarrow c = 10 - (m+3)^2$$

similarly

$$mx + c = 2 - \frac{x^2}{4} \Rightarrow \text{Disc} = 0 \Rightarrow c = 2 + m^2$$

$$\text{Now } m_1 + m_2 = -3$$

133. Since one unit towards negative direction X-axis

$$\rightarrow f(x+1)$$

134. On X-axis, one unit forward (a head)

$$135. \quad -1 \leq 2x \leq 3 \rightarrow \frac{-1}{2} \leq x \leq \frac{3}{2}$$

$$136. \quad \therefore -1 \leq \frac{x}{2} \leq 3 \rightarrow -2 \leq x \leq 6$$

137. Given that

$$-1 \leq f(x) \leq -2 \rightarrow -2 \leq 2f(x) \leq 4$$

in the same domain

$$138. \quad \therefore -1 \leq f(x) \leq 2 \Rightarrow 1 \leq f(x) + 2 \leq 2$$

$$139. \quad \therefore -1 \leq -x \leq 3 \Rightarrow -3 \leq x \leq 1$$

$$140. \quad f(-x) = f(x) \Rightarrow \text{symmetrical about the Y-axis}$$

$$\Rightarrow y = f(|x|)$$

$$141. \quad \text{Symmetry about the Y-axis and no part is below the X-axis} \Rightarrow y = |f(|x|)|$$

$$142. \quad \text{Symmetry about the Y-axis and no part is below the X-axis} \Rightarrow y = |f(|x|)|$$

143. Symmetry about the X-axis

$$144. \quad \sum_{r=1}^{1995} f\left(\frac{r}{1996}\right) = f\left(\frac{1}{1996}\right) + f\left(\frac{2}{1996}\right)$$

$$+ \dots + f\left(\frac{1995}{1996}\right)$$

$$= \left\{ f\left(\frac{1}{1996}\right) + f\left(\frac{1995}{1996}\right) \right\}$$

$$+ \left\{ f\left(\frac{2}{1996}\right) + f\left(\frac{1994}{1996}\right) \right\}$$

$$+ \dots + \left\{ f\left(\frac{997}{1996}\right) + f\left(\frac{999}{1998}\right) \right\} + f\left(\frac{998}{1996}\right)$$

$$(1+1+\dots+997 \text{ times})$$

$$+ f\left(\frac{1}{2}\right) = 997 + \frac{1}{2} = 997.5$$

$$\text{Now } [997.5] = 997$$

$$145. \sum_{r=1}^{1996} f\left(\frac{r}{1997}\right) = 1+1+\dots+998 \text{ times}$$

$$= 998$$

$$146. g_5(x) + g_5(1-x) = f(x) + f(1-x) = 1$$

$$147. \sum_{r=1}^{2n-1} 2 \cdot f\left(\frac{r}{2n}\right)$$

$$= 2 \left\{ f\left(\frac{1}{2n}\right) + f\left(\frac{2}{2n}\right) + f\left(\frac{3}{2n}\right) + \dots + f\left(\frac{2n-1}{2n}\right) \right\}$$

$$2 \left\{ 1+1+\dots+(n-1) \text{ times} + f\left(\frac{1}{2}\right) \right\}$$

$$= 2 \left\{ (n-1) + \frac{1}{2} \right\} = 2n-1$$

$$148. \sum_{r=0}^{2n} f\left(\frac{r}{2n+1}\right)$$

$$= f(0) + \sum_{r=1}^{2n} f\left(\frac{r}{2n+1}\right) = \frac{1}{1+\sqrt{a}} + n$$

$$= \frac{1}{1+\sqrt{a}} + 987 \rightarrow n = 987$$

MATRIX - MATCHING QUESTIONS

$$149. a \Rightarrow f(x) = \log_3(5+4x-x^2)$$

$$= \log_3(9-(x-2)^2)$$

Now $-\infty < 9-(x-2)^2 \leq 9$ but for $f(x)$ to get defined

$$0 < 9-(x-2)^2 \leq 9$$

$$\Rightarrow -\infty < \log_3(9-(x-2)^2) \leq \log_3 9$$

$$\Rightarrow \text{range}(-\infty, 2]^3$$

$$b \Rightarrow f(x) = \log_3(x^2 - 4x + 5)$$

$$= \log_3((x-2)^2 - 9)$$

$$\text{but here } 0 < (x-2)^2 - 9 < \infty, -\infty < f(x) < \infty$$

$$c \Rightarrow \log_3(x^2 - 4x + 5)$$

$$= \log_3((x-2)^2 + 1) \text{ but } (x-2)^2 + 1 \in [1, \infty)$$

$$\text{range} = [0, \infty)$$

$$d \Rightarrow \log_3(4x - 5 - x^2) \text{ but } -x^2 + 4x - 5 \text{ is always Negative}$$

$$D \text{ is } c < 0 \text{ and leading coeff} < 0$$

$$150. \text{ Let } f(x) = \frac{(x-5)(x-1)}{(x-2)(x-3)} \text{ if } -1 < x < 1 \text{ then}$$

$$f(x) = \frac{(-ve)(-ve)}{(-ve)(-ve)} = +ve \rightarrow p, r, s$$

$$\text{If } 1 < x < 2 \rightarrow f(x) < 0 \rightarrow q$$

$$\text{If } 3 < x < 5 \rightarrow f(x) < 0 \rightarrow q$$

$$\text{If } x > 5 \rightarrow f(x) > 0 \rightarrow p, r, s$$

ASSERTION - REASON QUESTIONS

$$151. f(\theta) = \frac{4 \sin^2 \theta \cos \theta - \cos \theta + \sin \theta}{4(\cos \theta + \sin \theta)}$$

$$= \frac{1}{2} \sin \theta (\sin \theta + \cos \theta) - \frac{1}{4}$$

$$= \frac{1}{4} [\sin(2\theta) - \cos(2\theta)] \in \left[\frac{-1}{2\sqrt{2}}, \frac{1}{2\sqrt{2}} \right]$$

$$152. f(x) \text{ is odd} \Rightarrow f(-x) = -f(x)$$

$$g(x) \text{ is odd} \Rightarrow g(-x) = -g(x)$$

$$F(x) = f(x) + g(x)$$

$$F(-x) = f(-x) + g(-x)$$

$$= \pm f(x) \Rightarrow F(x) \text{ is neither even nor odd}$$

$$\text{clearly } S_2 \text{ is true}$$

$$153. \text{ Conceptual}$$

$$154. f(1) = 4 \Rightarrow f^{-1}(4) = 1$$

$$155. f(x) = \frac{a^x}{a^x + \sqrt{a}}$$

$$\Rightarrow f(1-x) = \frac{a^{1-x}}{a^{1-x} + \sqrt{a}}$$

$$\Rightarrow f(x) + f(1-x) = 1$$

$$\Rightarrow \sum 2f\left(\frac{r}{2n}\right) = 2n - 1$$

INTEGER QUESTIONS

156. replace 'x' by $\frac{1-x}{1+x}$, we have

$$f(x) \cdot f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$$

$$\Rightarrow f(x) = \pm x^n + 1$$

$$\rightarrow (x) = x^3 + 1 \quad (\because f(3) = 28)$$

$$\text{Now } \sum_{n=1}^{10} (f(n) - 1) = \sum_{n=1}^{10} n^3 = 325$$

$$\Rightarrow \frac{1}{605} (3025) = 5$$

157. $f^2(x) \cdot f\left(\frac{1-x}{1+x}\right) = x^3 \dots \dots (1)$

replace 'x' by $\frac{1-x}{1+x}$

$$f^2\left(\frac{1-x}{1+x}\right) f(x) = \left(\frac{1-x}{1+x}\right)^3 \dots \dots (2)$$

from (1) and (2)

$$f^3(x) = x^6 \left(\frac{1+x}{1-x}\right)^3$$

$$\Rightarrow f(x) = x^3 \left(\frac{1+x}{1-x}\right)$$

$$f(-2) = \frac{8}{3} \rightarrow |f(-2)| = 2$$

158. $(f(x) - 1)^3 = x^6$

$$\Rightarrow f(x) = x^2 + 1 \rightarrow f(0) = 1$$

159. If $f(x) = x^3 + \frac{1}{x^3} - 4\left(x^2 + \frac{1}{x^2}\right) + 13$

$$= \left(x + \frac{1}{x}\right)^3 - 3\left(x + \frac{1}{x}\right)^2 - 4\left[\left(x + \frac{1}{x}\right) - 2\right] + 13$$

$$\Rightarrow f(2 + \sqrt{3}) = 9$$

SUBJECTIVE QUESTIONS

160. Given that $f(x+y) = f(x)f(y) \quad \forall x, y \in \mathbb{N}$

and $f(1) = 2$

To find 'a' such that,

$$\sum_{k=1}^n f(a+k) = 16(2^n - 1) \dots (1)$$

For this we start with

$$f(1) = 2 \dots (2)$$

then

$$f(2) = f(1+1) = f(1)f(1)$$

$$\Rightarrow f(2) = 2^2$$

Using (2)

Similarly we get,

$$f(3) = 2^3, f(4) = 2^4, \dots, f(n) = 2^n$$

Now equation (1) can be written as

$$f(a+1) + f(a+2) + f(a+3) + \dots + f(A)f(n) = 16(2^n - 1)$$

$$\Rightarrow f(A)f(1) + f(A)f(2) + f(A)f(3) + \dots + f(A)$$

$$f(n) = 16[2^n - 1]$$

$$\Rightarrow f(A)[f(1) + f(2) + f(3) + \dots + f(n)]$$

$$= 16[2^n - 1]$$

$$\Rightarrow f(A)[2 + 2^2 + 2^3 + \dots + 2^n] = 16[2^n - 1]$$

$$\Rightarrow f(A) \left[\frac{2(2^n - 1)}{2 - 1} \right] = 16[2^n - 1]$$

$$\Rightarrow f(A) = 8 = 2^3 = f(3) \Rightarrow a = 3.$$

161. Given that $4\{x\} = x + [x]$

where $[x]$ = greatest integer $\leq x$

$\{x\}$ = fractional part of x

$$\therefore x = [x] + \{x\}, \text{ for any } x \in \mathbb{R}$$

$\therefore$ Given equation becomes,

$$4\{x\} = [x] + \{x\} + [x]$$

$$\Rightarrow 3\{x\} = 2[x]$$

$$\Rightarrow [x] = \frac{3}{2} \{x\}$$

$$\text{Now } -1 \leq \{x\} < 1 \dots (1)$$

$$\Rightarrow 0 \leq \frac{3}{2} \{x\} < \frac{3}{2}$$

$$\Rightarrow 0 \leq [x] < \frac{3}{2} \text{ [Using equation (1)]}$$

$$\Rightarrow [x] = 0, [x] = -1$$

$$\text{If } [x] = -0$$

$$\Rightarrow \frac{3}{2} \{x\} = 0 \text{ [Using equation (1)]}$$

$$\Rightarrow \{x\} = 0$$

$$\therefore x = 0 + 0 = 0$$

$$\text{If } [x] = 1$$

$$\text{then } \frac{3}{2} \{x\} = 1 \text{ [Using equation (1)]}$$

$$\Rightarrow \{x\} = 2/3$$

$$\Rightarrow x = 1 + 2/3 = 5/3$$

$$\text{Thus, } x = 0, 5/3.$$

$$162. \text{ Let } f(x) = \frac{x+2}{x^2-8x-4}$$

$$\text{Domain of } f(x) \text{ is } x^2 - 8x - 4 \neq 0$$

$$\Rightarrow x \neq (4 \pm 2\sqrt{5})$$

$$\text{Range : Let } y = \frac{x+2}{x^2-8x-4}$$

$$x^2y - 8xy - 4y = x + 2$$

$$x^2y - x(8y+1) - (4y+2) = 0$$

$$\text{Let (i) } y = 0 \Rightarrow x = -2$$

$$\text{(ii) } y \neq 0 \Rightarrow (8y+1)^2 + 4y(2+4y) \geq 0$$

$$80y^2 + 24y + 1 \geq 0$$

$$\Rightarrow y \in \left(-\infty, \frac{-1}{4}\right] \cup \left[\frac{-1}{20}, \infty\right).$$

$$163. \text{ We have } f(x) = \frac{\alpha x^2 + 6x - 8}{\alpha + 6x - 8x^2}$$

$$\text{Let } k \in \mathbb{R} \text{ and } f(x) = k.$$

$$\Rightarrow \frac{\alpha x^2 + 6x - 8}{\alpha + 6x - 8x^2} = k$$

$$\Rightarrow \alpha x^2 + 6x - 8 = k(\alpha + 6x - 8x^2)$$

$$\Rightarrow (a + 8k)x^2 + (6 - 6k)x - 8 - ka = 0$$

$$\Rightarrow (6 - 6k)^2 - 4(a + 8k)(-8 - ka) \geq 0$$

$$(\because x \text{ is real})$$

$$\Rightarrow 36(1+k^2-2k)-4(-8a-64k-ka^2-8k^2) \geq 0$$

$$\Rightarrow 9 + 9k^2 - 18k + 8a + 64k + ka^2 + 8k^2 a \geq 0$$

$$\Rightarrow (8a + 9)k^2 + (a^2 + 46)k + (8a + 9) \geq 0$$

Since k is a real number, we have

$$8a + 9 > 0 \text{ and } (a^2 + 46)^2 - 4(8a + 9)^2 \leq 0$$

$$\Rightarrow a > -9/8 \text{ and } (a^2 + 46 - 2(8a + 9))(a^2 + 46 + 2(8a + 9)) \leq 0$$

$$\Rightarrow a > -9/8 \text{ and } (a^2 - 16a + 28)(a^2 + 16a + 64) \leq 0$$

$$\Rightarrow a > -9/8 \text{ and } (a - 2)(a - 14)(a + 8)$$

$$(a + 8) \leq 0$$

$$\Rightarrow a > -9/8 \text{ and } (a - 2)(a - 14) \leq 0$$

$$(\because (a + 8)^2 \geq 0)$$

$$\Rightarrow a > -8/9 \text{ and } 2 \leq a \leq 14$$

$$\Rightarrow 2 \leq 14.$$

$$f \text{ is onto if } 2 \leq a \leq 14$$

$$\text{Let } a = 3.$$

$$f(x) = \frac{3x^2 + 6x - 8}{3 + 6x - x^2}$$

$$0 \in \mathbb{R} \text{ and } f(x) = 0 \Rightarrow 3x^2 + 6x - 8 = 0$$

$$\Rightarrow x = \frac{-6 \pm 2\sqrt{33}}{6} = \frac{-3 \pm \sqrt{33}}{3} \in \mathbb{R}$$

f is not one-one.

$$164. \text{ Suppose } f(x) = Ax^2 + Bx + c \text{ is an integer whenever } x \text{ is an integer.}$$

$$\therefore f(0), f(1), f(-1) \text{ are integers}$$

$$\Rightarrow C, A + B + C, A - B \text{ are integers.}$$

$$\Rightarrow C, A + B, A - B \text{ are integers}$$

$$\Rightarrow C, A + B, (A + B) - (A - B) = 2A \text{ are integers.}$$

$$\text{Conversely suppose } 2A, A + B \text{ and } C \text{ are integers,}$$

$$\text{Let } n \text{ be any integer.}$$

$$\text{We have}$$

$$f(n) = An^2 + Bn + C = 2A$$

$$\left[\frac{n(n-1)}{2} \right] + (A+B)n + C$$

$$\text{Since } n \text{ is an integer, } n(n-1)/2 \text{ is an integer.}$$

$$\text{Also } 2A, A + B \text{ and } C \text{ are integers.}$$

$$\text{We get } f(n) \text{ is an integer for all integer } n.$$